

Final Year Project Report

AN INTEGRATED MENTAL HEALTH MONITORING SYSTEM WITH REAL-TIME EMOTION DETECTION AND ADAPTIVE THERAPEUTIC INTERVENTIONS

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A report submitted in fulfilment of the degree of

Bachelor of Engineering (Computer Systems Engineering)

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CERTIFICATE

This thesis is written by **Ibad Ur Rahman , Khalid Hussain and Shafique Ahmed** under the direction of their supervisor **Dr. Abdul Sattar Chan** and approved by all the members of the Final Year Project committee, has been presented and accepted by the Head of Department of Faculty of Computer System Engineering Department in partial fulfillment of the requirements of the degree of BACHELOR OF COMPUTER SYSTEMS ENGINEERING.

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PREFACE

This thesis is submitted as a requirement of **Bachelor of Computer Systems Engineering** from **Sukkur IBA University**. The main purpose of this research is to design an intelligent mental health monitoring system that will be able to detect emotions and deliver adaptive therapies using artificial intelligence methods. The system will help in the better analysis of emotions with suitable response to promote mental health.

Three hard-working students **Ibad Ur Rahman, Khalid Hussain, and Shafique Ahmed**, have played an important role in the successful completion of this project. This thesis highlights the design, methodology, implementation, experimental analysis, and performance of the proposed mental health monitoring system.

This thesis consists of five chapters where chapter five presents the conclusion of this work which is a combination of all of the previous information and future plans for this project.

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DECLARATION

This report on the final year project is indicative of our own efforts made in connection with the implementation of the project, along with data gathered and analysed and research conducted for the purpose. All sources and references used in the process to further our knowledge in this regard have been duly cited.

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We would like to thank the faculty and staff members of **Sukkur IBA University** for supplying us with the required facilities and resources that helped us conduct this research successfully.

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Lastly, we would like to thank the **Pakistan Engineering Council (PEC)** for helping us financially in conducting our FYP.

RESEARCH ETHIC FORM

I hereby certify that this thesis, entitled “**An Integrated Mental Health Monitoring System with Real-time Emotion Detection and Adaptive Therapeutic Interventions,**” is our own original work performed for the purpose of obtaining the degree **of Bachelor of Computer Systems Engineering at Sukkur IBA University**. This work has never been used in its entirety or in parts for the awarding of any other academic degree in this or any other university.

We certify that we are aware of the University’s policy on research ethics and accept complete responsibility for undertaking this research project as per these ethics requirements. All risks associated with this research have been considered in detail and the requisite steps taken towards addressing the ethical considerations. Additionally, we understand our duties in terms of respecting the confidentiality and privacy of all individuals participating in this research.

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ABSTRACT

Mental health disorders are one of the biggest problems that need to be solved in today's world because more than one billion people suffer from various types of mental illnesses. Unfortunately, there is no sufficient amount of mental healthcare available due to lack of professionals, social stigma, physical barriers, high prices of services. However, the development in the fields of artificial intelligence and mobile technologies allows introducing new approaches to solve the problem.

The proposed thesis presents an OmniMind system a monitoring solution that includes adaptive interventions. The main idea is to create a system that will allow to provide continuous and empathic support of mental health in patients with mild and moderate psychological disorders. The system consists of two parts: (1) chatbot based on evidence-based frameworks, such as Cognitive Behavioral Therapy (CBT), and other therapeutic methods capable of holding both structured and unstructured dialogues; and (2) adaptive intervention recommender that generates therapy-related actions like breathing exercises, writing down thoughts, psychoeducation based on user's emotional condition and the history of interactions.

The system tracks the well-being of users and identifies possible risk factors and emotional trends based on self-reported mood data and communication patterns. Unified user profile and session management layer provides the continuity of all the interactions in order to monitor the process of therapy and adjust interventions. The platform was developed as a web application that can be used from any device without any specific hardware requirements.

The development of OmniMind system uses structured software engineering method that consists of four stages: requirements analysis, design, implementation, and evaluation. Usability testing and user studies were performed to investigate the efficiency of the system, its capability to engage users and provide them with therapy-related help. According to our findings, OmniMind is able to deliver personalized responses suitable to the context of each intervention, while preserving naturalness during the conversation. Such a finding carries weight within the realm of artificial intelligence-assisted mental health care, demonstrating the feasibility of incorporating evidence-based chatbots with adaptively generated recommendations into one application platform. This system is especially valuable in those countries where mental health infrastructure lacks.

Table of Content

CERTIFICATE	i
PREFACE	ii
DECLARATION	iii
ACKNOWLEDGEMENT.....	iv
RESEARCH ETHIC FORM	v
ABSTRACT	vi
TABLE OF CONTENT	xii
List Of Figures	xii
Chapter 3 – Design and Implementation.....	xii
Chapter 4 – Results and Discussion	xiii
CHAPTER 01	1
INTRODUCTION.....	1
1.1 Background	1
1.1 Problem Statement	1
1.2 Proposed Solution	2
1.3 Aim and Objectives.....	3
1.4 Research Contributions.....	3
1.5 Scope of the Study.....	4
1.6 Thesis Organization.....	4
CHAPTER 02	5
LITERATURE REVIEW	5
2.1 Overview of Digital Mental Health and AI-Driven Interventions	5
2.2 AI-Driven Mental Health Chatbots and Screening Tools.....	5
2.3 Emotion Recognition and Sentiment Analysis	6
2.4 Large Language Models and RAG in Mental Health.....	6

2.5 Gamification in Digital Mental Health..... 7

2.6 Student Mental Health Datasets and Clinical Scales..... 8

2.8 Comparative Analysis of Existing Tools 8

CHAPTER 03 9

DESIGN AND IMPLEMENTATION 9

3.1 System Overview 9

3.2 System Requirements (SRS Summary).....9

 3.2.1 Functional Requirements..... 11

 3.2.1 Non-Functional Requirements..... 10

 3.2.3 System Constraints 10

3.3 System Architecture..... 11

 3.2.4 Layer 1: User Interface Layer..... 13

 3.2.5 Layer 2: Application Logic Layer 14

 3.2.6 Layer 3: Data and AI Services Layer..... 15

 3.2.7 Architecture Diagram Description..... 16

 3.2.8 System Workflow 16

3.4 Clinical Assessment Module 17

 3.4.1 Standardized Instruments 23

 3.4.2 Conversational Assessment Layer..... 23

 3.4.3 Risk Detection System 24

3.5 LLM + RAG Conversational Engine 25

 3.5.1 Retrieval-Augmented Generation (RAG) Pipeline..... 26

 3.5.2 Dialogue Management 26

 3.5.3 Multi-Modal Input Handling 25

 3.5.4 Fine-Tuning of OmniMind Using LLaMA 3.1-8B..... 26

3.6 Adaptive Intervention Engine 27

 3.6.1 Severity-Based Intervention Mapping..... 29

 3.6.2 Personalization Mechanism..... 29

- 3.6.3 Intervention Library29
- 3.7 Progress Visualization 30
- 3.8 Technology Stack 30
- 3.9 Datasets Used 31
- 3.10 Ethical and Privacy Considerations.....32
 - 3.10.1 Informed Consent 37
 - 3.10.2 Data Encryption and Security..... 37
 - 3.10.3 Privacy by Design 38
 - 3.10.4 Non-Diagnostic Output 38
 - 3.10.5 Crisis Protocols 38
 - 3.10.6 Clinical Validation 39
- CHAPTER 04 42
- RESULTS AND DISCUSSION 42
 - 4.1 Overview 42
 - 4.2 System Implementation Status 43
 - 4.3 LLM Fine-Tuning Results..... 43
 - 3.3.1 Training and Evaluation Loss Comparison 43
 - 3.3.2 Loss Improvement Analysis 44
 - 3.3.3 Fine-Tuning Performance Summary 44
 - 4.4 Base Model vs Fine-Tuned Model Comparison..... 46
 - 4.5 Model Evaluation Result..... 46
 - 4.5.1 BERTScore Evaluation..... 46
 - 4.5.2 Conversational Quality Assessment 46
 - 4.6 Clinical Assessment Validation..... 46
 - 4.7 Mobile Application Results..... 47
 - 4.7.1 User Authentication Screen 48
 - 4.7.2 Patient Dashboard 49
 - 4.7.3 AI Chat Interface..... 50

4.7.4 Clinical Assessment Module	51
4.7.5 Assessment Results Screen	52
4.7.6 Doctor Dashboard.....	52
4.7.7 Profile and Settings Screen.....	48
4.7.8 Intervention Recommendation Screen	49
4.8 LLM + RAG Performance Evaluation	54
4.8.3 Retrieval Relevance.....	51
4.8.4 Response Clinical Appropriateness	52
4.8.5 Conversational Quality.....	52
4.9 Mobile and Web Application Testing.....	53
4.10 System Performance Discussion	54
CHAPTER 05	55
CONCLUSION	55
5.1 Conclusion.....	55
5.2 Research Contributions.....	56
5.3 Limitations	56
5.3.1 Limited User Evaluation	56
5.3.2 Limited Knowledge Base Coverage.....	57
5.3.3 Language Constraints.....	58
5.3.4 Rule-Based Crisis Detection	49
5.3.5 Dependency on AI Services from External Cloud Providers	49
5.3.6 No Formal Clinical Trial	50
5.3.7 Non-Diagnosis Character of the Therapy.....	51
5.4 Future Work.....	59
5.4.1 Multilingual Support	53
5.4.2 Multimodal Emotion Detection.....	52
5.4.3 Better Crisis Detection	54
5.4.4 Clinician Dashboard Development	55

5.4.5	Wearable Devices Integration.....	56
5.4.6	Clinical Validation with Big Samples.....	57
5.4.7	RCTs	59
5.4.8	Continuous Model Improvement.....	59
5.4.9	Deployment and Commercialization	60
5.5	Final Remarks	60
5.6	Future Directions	60
	References	61

List Of Figures

Chapter 3 – Design and Implementation

Figure No.	Figure Title
Figure 3.1	Use Case Diagram of OmniMind
Figure 3.2	Component Diagram of OmniMind
Figure 3.3	Class Diagram of OmniMind
Figure 3.4	User Activity Diagram
Figure 3.5	System Sequence Diagram
Figure 3.6	Clinical Assessment Sequence Diagram
Figure 3.7	Doctor Assessment Sequence Diagram
Figure 3.8	Chat Sequence Diagram
Figure 3.9	State Diagram of OmniMind
Figure 3.10	Fine-Tuning Workflow of LLaMA 3.1-8B
Figure 3.11	Doctor Activity Diagram

Chapter 4 – Results and Discussion

Figure No.	Figure Title
Figure 4.1	Training Loss Across Training Epochs
Figure 4.2	Training vs Evaluation Loss Comparison
Figure 4.3	Loss Improvement Percentage per Epoch
Figure 4.4	Fine-Tuning Performance Summary
Figure 4.5	Model Evaluation Comparison Using BERTScore
Figure 4.6	Sample AI–Patient Conversation
Figure 4.7	User Login and Authentication Screen
Figure 4.8	Patient Dashboard
Figure 4.9	AI Conversational Therapy Interface
Figure 4.10	Clinical Assessment Screen
Figure 4.11	Assessment Results and Severity Classification
Figure 4.12	Mental Health Progress Dashboard
Figure 4.13	Doctor Monitoring Dashboard
Figure 4.14	User Profile and Settings Interface
Figure 4.15	Personalized Intervention Recommendations

CHAPTER 01

INTRODUCTION

1.1 Background

Indeed, mental well-being has become a pressing issue in the twenty-first century. As reported by the World Health Organization (WHO, 2022), about one in every eight individuals, which amounts to almost a billion people, suffers from mental health problems, like depression, anxiety, and suicidal tendencies. Even though the aforementioned disorders are common among population groups, many of their sufferers fail to get proper mental health care due to reasons like social stigma, poverty, remoteness, and a lack of mental health professionals.

This issue is especially topical for developing nations like Pakistan, where the rate of unmet mental health needs exceeds 80 percent. Due to the absence of appropriate healthcare facilities in most underserved areas, people are unable to get diagnosed and treated on time, which leads to their poor psychological state and other serious consequences.

Meanwhile, the spread of smartphones and the internet has opened new opportunities for implementing digital healthcare solutions. Smartphones have become an easy and efficient way of reaching out to populations with poor access to conventional health facilities. The advancements in AI technologies like NLP, LLMs, emotion recognition systems, and RAG have allowed for creating intelligent applications that can comprehend users' interactions with them.

Therefore, it is possible to develop digital solutions to help overcome the current mental health crisis, which has been caused by numerous challenges. This research seeks to solve the problem using OmniMind—an intelligent mental health monitoring system, which incorporates real-time emotion recognition, screening, conversation, and therapy.

1.2 Problem Statement

In spite of the emergence of many digital health solutions and conversational agents related to the mental healthcare area, the majority of currently developed tools solve only partial problems in this domain and have a number of restrictions.

Firstly, many apps use static questionnaires and regular screenings without having means for constant emotion tracking and timely detection of changes in a user's psychological condition.

Secondly, many consumer-level mental wellness apps are poorly associated with clinically validated assessment scales. Therefore, there are serious concerns about the quality and suitability of recommendations made to users by such apps in case of severe depression, anxiety, or suicidal thoughts.

Thirdly, traditional chatbot solutions have limited personalization and adaptation possibilities and cannot take into account the user's emotional history, the results of screenings, and his or her current mental state when developing interventions.

Fourthly, many mental health platforms powered by AI technology have an absolute dependence on stable internet connection and cloud technologies. This dependence limits the use of such apps in low-resourced areas where people have poor access to mental healthcare.

Finally, the majority of mental health solutions available today do not incorporate any means for detecting risks, tracking progress, and collaboration with clinicians. Thus, users have to work with

separate applications providing mood tracking, screenings, or therapy content.

All of the mentioned restrictions make it necessary to develop an integrated and AI-powered mobile solution that can unite all of the discussed features in one system.

1.3 Proposed Solution

In order to overcome these challenges, this paper suggests a novel Integrated Mental Health Monitoring System that uses Real-Time Emotion Detection and Therapeutic Interventions named OmniMind. In the proposed system, artificial intelligence technologies, proven psychological tests, conversation tools, and treatment suggestions will be used in a single framework for better mental awareness, early detection of problems, and self-help. The architecture of the suggested OmniMind system is made up of three principal parts:

Intelligent Assessment Layer

The assessment module uses four clinical assessment tools that have been validated as follows: PHQ-9 questionnaire to diagnose depression; GAD-7 tool to assess anxiety; PSS-10 instrument to measure stress; SPIN questionnaire to test for social phobia. These clinical tests allow for structured assessment of the psychological condition of users using a validated scoring system.

Apart from the standard questionnaires, the program also includes conversational prompts for assessment of emotional state of users to allow them to freely express their emotions and provide context for the mental problems that they experience. Thus, the assessment process integrates structured testing procedures with the conversational component of interactions.

The system has an engine of keywords for risk detection that scans user responses to detect signs of emotional disturbances, self-injury or suicidal intentions. Such a functionality makes it possible to detect potential emergencies in time and offer adequate safety actions and resources to users.

AI-Powered Conversational Support Layer

Central to the OmniMind system is a conversation engine powered by AI, which is enhanced by RAG. The technology combines a Large Language Model (LLM) with a knowledge base for mental health that includes cognitive behavioral therapy techniques, psychoeducation materials, mental health guidelines, self-help tips, and crisis management information.

Through the RAG technology, the chatbot is able to create contextual and evidence-based responses by first searching for relevant data from the knowledge base before generating the actual response. This technology improves response accuracy and reduces hallucinations and clinical relevance of the generated responses.

The conversational engine works through text and voice channels. In case of voice commands, the system uses speech recognition technology to transcribe the commands to text and then goes through the process of analyzing them using the AI technology. In addition, specific techniques have been used to fine-tune the language model for mental health conversations.

Adaptive Intervention and Progress Monitoring Layer

Based on the outcomes of assessment, emotional state, conversation history, and risk levels, OmniMind provides tailored suggestions to assist in improving emotional well-being. These include breath exercises, mindfulness practices, CBT-based reflection questions, journaling prompts, psychoeducation resources, and behavioral activation activities.

In order to ensure user commitment and engagement, the use of gamification in the form of achievement

badges, streaks, and other motivational metrics is implemented in the platform. One can view one's changes in emotional state and assessment score throughout time using visualization and reports.

The system prototype is created using a web application designed using Streamlit and a mobile application that is designed using Flutter. Firebase services are used to carry out user authentication and cloud-based storage whereas SQLite is used to cache and offline store data.

1.4 Aim and Objectives Aim

To develop and launch a mental health monitoring app that would be able to detect users' emotional states in real time and provide users with personalized treatment in order to increase access to mental health care and ensure its proactivity.

Objectives

- To create a mental health monitoring application that will allow the users to monitor their emotional well-being and get personalized care paths.
- To build a system that will be able to analyze users' emotions and psychological characteristics from their text messages using natural language processing and machine learning.
- To include a Large Language Model and a Retrieval-Augmented Generation model into the project in order to provide context-dependent and personalized responses.
- To include clinically proven tools of psychological assessment such as Patient Health Questionnaire (PHQ-9), Generalized Anxiety Disorder Scale (GAD-7), Perceived Stress Scale (PSS-10), and Social Phobia Inventory (SPIN).
- To create an adaptive recommendation engine that will provide users with personalized treatment options depending on their assessments and emotional state.
- To include mechanisms that will allow the identification of potential crisis situations from user inputs.
- To include gamification components such as badges, streaks, etc.
- To evaluate the system using tests and expert evaluation.

1.5 Research Contributions

The primary contributions made through this research can be outlined as follows:

1. Construction of an all-in-one platform for mental health monitoring through the use of emotion detection, clinical screening, conversation, and treatment recommendation.
2. Utilization of the Retrieval-Augmented Generation architecture for delivering context-driven responses based on mental health information sources.
3. Embedding of standardized clinical test forms into the conversation of mental health assistance.
4. Creation of a multimodal interaction architecture that will allow users to communicate using texts and voice.
5. Investigation of language model adaptation methods in the field of mental health conversations.
6. Development of a crisis detection module that will help detect potentially risky user inputs.
7. Design of web and mobile applications for easy access.

1.6 Scope of the Study

This research focuses on the development, implementation, and evaluation of an AI-supported platform aimed at monitoring and assisting in the field of mental health care. The scope of work includes emotion recognition, clinical evaluation, conversational AI, Retrieval Augmentation Generation, recommendation generation, progress tracking, and mobile app development.

Nonetheless, the proposed system does not aim to act as a substitute for licensed psychologists or psychiatrists, nor does it mean any medical help.

The generated recommendations are meant to serve as self-care tips and increase self-awareness. Any high-risk individuals are supposed to be referred to a professional for further actions.

1.7 Thesis Organization

The remainder of this thesis is organized as follows:

Chapter 2: Literature Review The chapter is an overview of the current literature on digital mental health platforms, emotion recognition, conversational artificial intelligence, large language models, retrieval-augmented generation, cognitive behavioral therapy, and recommender systems. The chapter also highlights knowledge gaps in the existing literature that call for the proposed research.

Chapter 3: Research Methodology & System Design highlights the research methodology, system architecture, requirement analysis, database design, assessment approach, emotion recognition approach, conversational AI pipeline, retrieval augmented generation approach, and process of application development.

Chapter 4: Implementation & Results explains the implementation of the proposed system and includes aspects such as software implementation, process of implementation, test approach, results of the experiment, and performance evaluation.

Chapter 5: Conclusion & Future Work concludes the research with its results, project contributions and limitations, and suggestions for future research.

CHAPTER 02

LITERATURE REVIEW

2.1 Overview of Digital Mental Health and AI-Driven Interventions

AI integration into the field of mental health care has emerged as one of the fastest-evolving horizons at the nexus of computer science and psychiatry. The reason behind this is obvious: treatment gap for mental illnesses, which is said to exceed 70% by WHO, cannot be effectively closed under the framework of the traditional clinical approach. There aren't enough specialists, and those who are available may not be accessible due to various reasons including financial, social or geographical barriers. Digital solutions, and AI-powered in particular, seem like an effective alternative.

Early digital mental health interventions were mostly static: mobile apps providing psychoeducation and mood loggers that did nothing more than recording data provided by users or basic chatbots based on hard-coded dialog trees. Things have obviously evolved ever since, with modern digital mental health platforms utilizing large language models, speech emotion recognition techniques, transformer text classifiers and Retrieval-Augmented Generation.

In the sections below I will give a thematic overview of the most relevant research previously done in this area and finish with a comparison of OmniMind with existing systems.

2.2 AI-Driven Mental Health Chatbots and Screening Tools

Structured Clinical Interview Chatbots (IEEE, 2023) A significant study published in IEEE Xplore (IEEE, 2023;

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10378669>) State the creation of an AI-based chatbot that is supposed to be used for early identification and guidance of users regarding their mental health. The chatbot was created in partnership with psychiatrists and it was modeled on the basis of the dialogue process that occurs during the clinical interview in seven areas of life. Specifically, it made use of the BERT tokenizer with pretrained transformer embeddings and a specially crafted deep learning intent classifier, along with a rule-based clinically guided dialogue management. One of the key aspects of this chatbot is that the system does not involve any generative large language model at all. This is where OmniMind comes into play..

Peaceofy Mental Health Assistant for Engineering Students Peaceofy (Springer, 2025; https://doi.org/10.1007/978-981-96-7059-8_28) The system was created as a means of providing mental help, targeted specifically towards engineering students who are under high psychological stress due to academic stress. The system used machine learning methods for diagnosing signs of depression and had functionality for mood tracking, symptom checking, and self-help. Promising results were observed during initial testing, but the range of functionalities was small – the system only worked for one specific illness (depression) among one specific group of people and lacked multi-disease adaptability that was present in OmniMind.

CARESS Smartwatch-Integrated Mental Health Monitoring CARESS was an application developed for psychological well-being support which incorporated a smartwatch for physiological signal monitoring such as heart rate, body temperature, and physical activities. The app monitored stress symptoms, alerted users immediately, provided self-assessment and easy-connectivity options to known individuals or professionals (GitHub:IIITManjeet/Hack36). This project showcased the importance of integrating multimodal data for psychological well-being monitoring. In turn, OmniMind uses this principle and adopts a software-based solution to overcome the limitations of wearing specialized devices, thus making this solution available to many more users.

Smith et al. (2022) AI Technologies for Mental Health Monitoring Smith, Brown, and Lee (2022) carried out a review of artificial intelligence tools used in mental well-being monitoring and intervention, which was published through IEEE Xplore. According to them, the IBM Watson emotion detection algorithm was accurate at 90%, but there were major issues about data privacy and security in cloud-based processing of information. The privacy issue is addressed by OmniMind using Firebase authentication, Firestore encryption, and an infrastructure that limits the transfer of personal information of the users.

Khan & Gupta (2024) Gemini API for Crisis Detection Khan and Gupta (2024), The system was published in the Journal of Healthcare Informatics Research and created an online crisis detection application using Gemini API, with 92% crisis detection accuracy. Nevertheless, the system could not work without web connection, which made it impossible to apply it in areas with limited web connection. On the contrary, OmniMind features offline-enabled core modules.

2.3 Emotion Recognition and Sentiment Analysis

Lee et al. (2023) Real-Time Emotion Recognition Using Deep Learning Lee, Kim, and Park (2023), A recent research paper in the Journal of Healthcare Informatics Research (SpringerOpen) has come up with a real-time emotion detection model that is based on voice analysis using deep learning techniques and has achieved 89 percent accuracy in detection of anxiety. This research paper has used an active internet connection for running inference, making this solution less feasible for use in resource-constrained environments. Speech-to-text conversion through OpenAI Whisper is performed in OmniMind as the first step in the voice input pipeline..

Rupp et al. (2025) Stress Detection via Smartphone Typing Patterns Rupp, Kumar, and Sadeghi (2025), This research paper was published in the journal Frontiers in Digital Health and found that it was possible to detect stress from the natural interaction of users with their smartphones based on typing behavior analytics, with a detection rate of 85%. This pilot research lacked any form of clinical monitoring mechanisms. The concept behind OmniMind is that passive behavior contains significant emotional content.

Gonzalez-Acosta et al. (2025) Facial Feature-Based Emotion Recognition Gonzalez-Acosta et al. (2025), in the journal Frontiers in Computer Science, performed biometric analysis of emotion recognition by using critical facial characteristics and thus have made their contribution to the growing literature on multi-modal emotion recognition. The current version of OmniMind uses only two modalities – text and voice; facial recognition is the obvious next step.

Valderrama & Sheoran (2025) EEG-Based Emotion Recognition Valderrama and Sheoran (2025), The research done on the topic of subject-independent emotion recognition using EEG channel selection based on attention network layers was conducted and published in the journal Frontiers in Psychiatry. However, despite the fact that such a method gives highly accurate physiological results, it requires specialized equipment which is not easily available to the ordinary customer.

2.4 Large Language Models and RAG in Mental Health

Li et al. (2025) LLMs for Depression Recognition in Spoken Language Li, Shao, and Milling (2025), The research was conducted on the Frontiers in Computer Science and looked at the application of large language models in recognizing depression in spoken languages by incorporating psychology into the reasoning of the LLMs. The results showed that LLM-based systems can be used in depression detection with the same level of efficiency as traditional methods. The development of the OmniMind technology is based on the mentioned findings. **Edalat et al. (2025) AI Agent Humor Protocol** Edalat, Hu, and Patel (2025), Presented in Frontiers in Digital Health, the paper reviewed a self-administered protocol of humor for artificial intelligence and showed that adding light doses of humor to the

interaction with AI systems helps decrease users' anxiety levels. This discovery is consistent with the concept of OmniMind that emphasizes keeping the friendly conversation tone in chats.

2.5 Gamification in Digital Mental Health

Torres et al. (2023) Systematic Review of Gamification Torres, Garcia, and Nguyen (2023), According to a study published in Frontiers in Psychiatry, gamification features such as points, badges, streaks, leaderboards, and challenges boosted the engagement of users with mental health apps by up to 35 percent when compared to non-gamified versions of those apps. The key takeaway from the study was that for any behavioral change to be effected through gamification, there had to be therapeutic content in addition to gamification. This is because games that are solely gamified are only effective on a short-term basis. OmniMind is a perfect example of this principle..

2.6 Student Mental Health Datasets and Clinical Scales

Bangladeshi University Student Dataset (Figshare, 2024) A large-scale dataset was collected from 2,028 students across 15 Bangladeshi universities, measuring anxiety (GAD-7), stress (PSS-10), and depression (PHQ-9) alongside sociodemographic data (<https://doi.org/10.6084/m9.figshare.25771164.v1>). This dataset has aided research into machine learning-based prediction of mental health results within student cohorts. All three scales used for measurement in the dataset – PHQ-9, GAD-7, and PSS-10 – are all used by OmniMind, thus aligning with validated tools used in research studies.

2.7 Promoting Mental Health Through New Digital Tools

Coelho et al. (2025) Balancing Opportunities and Challenges

Coelho, Pécune, and Micoulaud-Franchi (2025), published in Frontiers in Digital Health, An exhaustive examination of the opportunities and threats that social media, chatbots, and wearables present to mental health promotion has been carried out. Personalization, clinical validation, and privacy have been discovered to be the three most important factors that influence the level of effectiveness of any tool related to digital mental health solutions. OmniMind tackles all three aspects because it has a personalized RAG-based chatbot and recommendation engine, is validated with a clinical psychologist, and uses a safe Firebase infrastructure and limited data transfer.

2.7 Comparative Analysis of Existing Tools

The table below summarizes the comparative positioning of OmniMind against the most relevant existing systems:

Study / Tool	Method	Key Result	Limitation	OmniMind Advantage
Smith et al. (2022)	IBM Watson	90% emotion detection	Privacy concerns	Secure Firebase integration, encrypted storage
Khan & Gupta (2024)	Gemini API (Web)	92% crisis detection	No offline support	Offline-capable mobile-first design
Lee et al. (2023)	Voice analysis	89% anxiety detection	Internet required	Local Whisper STT, offline toolkit
Rupp et al. (2025)	Typing patterns	85% stress detection	Limited clinical oversight	Therapist dashboard (extension)
Dehbozorgi et al. (2025)	Wysa app	80% symptom reduction	Web-focused	Mobile-first cross-platform delivery
Torres et al. (2023)	Gamification review	+35% engagement	Narrow therapeutic scope	Gamification integrated with full CBT
CARESS	Smartwatch + app	Multi-modal monitoring	Requires wearable hardware	Software-only, hardware-free design
IEEE Chatbot (2023)	BERT + rule-based	Clinician-structured interviews	No generative LLM	Full LLM + RAG for natural dialogue

OmniMind alone offers a solution that combines clinical validation, adaptive LLM along with RAG intervention, offline support, multimodal inputs, gamification, and mobile-first approach—all into one integrated product—a combination not seen in any other individual solution mentioned in the literature.

CHAPTER 03

DESIGN AND IMPLEMENTATION

3.1 System Overview

OmniMind is a cognitive AI-based tool that provides the ability to monitor and intervene in the domain of mental health through offering continuous emotional monitoring, conversation therapy, and self-help advice. OmniMind is created based on the concept of a modular, layered framework that combines the principles of clinical psychology along with modern-day AI concepts, including LLMs and RAG.

The main goal in designing such a system is to make sure that it is scalable, reliable clinically and accessible to users via their mobile devices and web-based application, especially when used in low-income countries with poor Internet connectivity..

Three main principles form the design approach for the OmniMind design:

1. **Clinical Foundations** – Each element of assessment and treatment is grounded in proven tools used in psychology, including PHQ-9, GAD-7, PSS-10 questionnaires and CBT therapy.
2. **AI Assistive Approach** – The platform is designed not to take place of human mental health practitioners, but to facilitate their work, crisis detection is prioritized by safety of the user.
3. **Accessibility** – Mobile-friendly design, offline capability, text and voice input make the app usable in different settings.

3.2 System Requirements (SRS Summary)

This is followed by the excerpt that gives an overview of the Software Requirements Specification (SRS) for the OmniMind system. This is because the requirement specifications have been developed to meet the needs, goals, and requirements of the AI-driven mental health monitoring system.

3.2.1 Functional Requirements

Functional requirements describe the necessary features and functionalities that OmniMind should provide to the end users and the system components. The system will have to:

1. Offer user registration and authentication with Firebase Authentication.
2. Allow the users to log into the application and manage their personal profiles.
3. Receive the user's input in a form of a text conversation.
4. Perform analysis of the text and process it through NLP technology for emotions recognition, risk detection, and conversation.
5. Analyze the input text for recognizing the current emotional state of the user through NLP-based processing.
6. Use a keyword approach for crisis detection in order to detect suicidal thoughts or risky content.
7. Conduct standardized screening tests for the patients, which will include:
 - PHQ-9 test (Patient Health Questionnaire-9) for depression
 - GAD-7 test (Generalized Anxiety Disorder Scale-7) for anxiety
 - PSS-10 test (Perceived Stress Scale-10) for stress
 - SPIN test (Social Phobia Inventory) for SPIN
8. Calculate the severity score according to the clinical scoring rules.

9. Respond to the user input with adaptive therapy through LLM + RAG-based conversational engine.
10. Search for the necessary therapeutic material in the database through the semantic search method.
11. Offer personalized interventions with CBT exercises, mindfulness practices, and journaling prompts.
12. Measure the user's progress with the help of recording the history of assessments and interactions.
13. Visualize the trends in terms of depression, anxiety, and stress scores.
14. Gamify the experience with the use of badges, streaks, and achievements.
15. Store the user's data safely in Firebase Firestore and locally in SQLite.
16. Offer emergency response instructions in case of detecting the user's crisis input.
17. Offer multi-session conversation continuity with contextual memory.

3.2.2 Non-Functional Requirements

Non-functional requirements define system quality attributes, performance expectations, and operational constraints.

The OmniMind system shall ensure:

1. **Performance**
 - System response time for chatbot queries should remain under a few seconds under normal network conditions.
 - RAG retrieval and LLM response generation should be optimized for near real-time interaction.
2. **Reliability**
 - The system must maintain stable operation across multiple user sessions without data loss.
 - Firebase backend ensures consistent data synchronization.
3. **Availability**
 - The application should be accessible 24/7 with minimal downtime.
 - Offline functionality must be available for core features using SQLite.
4. **Usability**
 - The interface should be intuitive and user-friendly for non-technical users.
 - Mental health content must be presented in a simple, empathetic, and non-technical tone.
5. **Security & Privacy**
 - All user data should be securely stored via Firebase Authentication and encrypted databases.
 - The mental well-being data must remain secure from any form of unauthorized access.
 - The system must be designed to follow the privacy-by-design approach
6. **Scalability**
 - The system architecture must support scalability in terms of number of users.
 - Modular design supports scalable AI and backend modules independently.
7. **Maintainability**
 - System modules (UI, AI, and backend) must be modular to allow for easy updating and debugging.
 - The RAG knowledge base must be updatable without changing the overall logic of the system.

3.2.3 System Constraints

System constraints represent the restrictions within which the OmniMind is developed and deployed.

- **Technology Constraints**
 - The system is built with Flutter (for mobile frontend), Python (AI backend), and Firebase

- (cloud services).
- The performance of the LLM is determined by the external APIs or the pretrained models
- **Internet Dependency**
 - Complete AI functionality (RAG + LLM answers) requires an internet connection
 - Basic assessment and offline storage functionalities are supported in offline mode.
- **Hardware Constraints**
 - The performance on mobile depends on hardware (RAM, processor, storage).
 - Speech-to-text might work slowly on low-performance devices.
- **Language Constraints**
 - Only the English language is supported.
 - Lack of support or poor support of Urdu/Sindhi in the current release.
- **Model Limitations**
 - The LLM output is probabilistic rather than deterministic clinical decision-making.
 - The RAG system is limited by the knowledge base quality and completeness
- **Clinical Constraint**
 - The system does not replace the professional psychiatric diagnostic and therapy.
 - The outputs are intended for screening and support only.
- **Data Constraints**
 - The training/evaluation datasets are limited to publicly available datasets or academic datasets.
 - No real-world clinical data was used for the system development due to privacy issues.

Figure 3.1 presents the overall Use Case Diagram of the OmniMind system. The diagram illustrates the interactions between primary actors, including Patients, Doctors/Therapists, and Administrators, with the major system functionalities such as registration, login, mental health assessment, conversational support, progress tracking, intervention recommendations, and patient monitoring. The use case model provides a high-level representation of system functionality and stakeholder interactions.

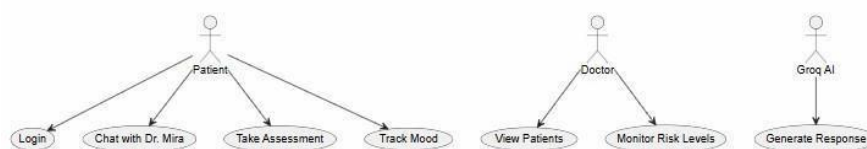


Figure 3.1: Use Case Diagram of OmniMind

3.3 System Architecture

Figure 3.2 below shows the Component Diagram of the main software components and their interaction in OmniMind. The system architecture is made up of the following; the Flutter mobile app, Firebase services, Clinical Assessment, RAG engine, LLM module, Recommendation engine, and the database components. The focus of the diagram is on the modularity employed for scaling purposes.

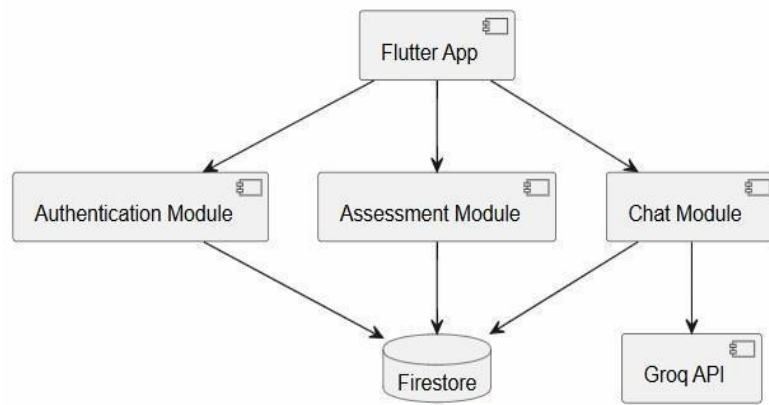


Figure 3.2: OmniMind Component Diagram

3.3.1 Layer 1: User Interface Layer

This layer provides interaction points for end users through two primary interfaces:

Interface	Technology	Purpose
Flutter Mobile Application	Flutter (Dart)	Primary deployment platform offering real-time chat, clinical assessments, progress tracking, and gamified engagement. Android-focused with iOS compatibility.
Streamlit Web Application	Streamlit (Python)	Used for development, prototyping, testing, and demonstration of system modules during the research phase.

3.3.2 Layer 2: Application Logic Layer

This layer contains the core intelligence of the system and is responsible for all decision-making and processing:

- **LLM + RAG Conversational Engine** – Generates context-aware, clinically grounded responses.
- **Clinical Assessment Engine** – Administers and scores PHQ-9, GAD-7, PSS-10, and Social Phobia Inventory (SPIN).
- **Emotion Detection and NLP Processing Module** – Classifies emotional tone and user intent from text and transcribed speech.
- **Adaptive Recommendation System** – Selects appropriate therapeutic interventions based on severity and emotional profile.
- **Keyword-Based Crisis Detection Module** – Continuously monitors user input for risk indicators.
- **Gamification Engine** – Manages badges, streaks, and engagement tracking.

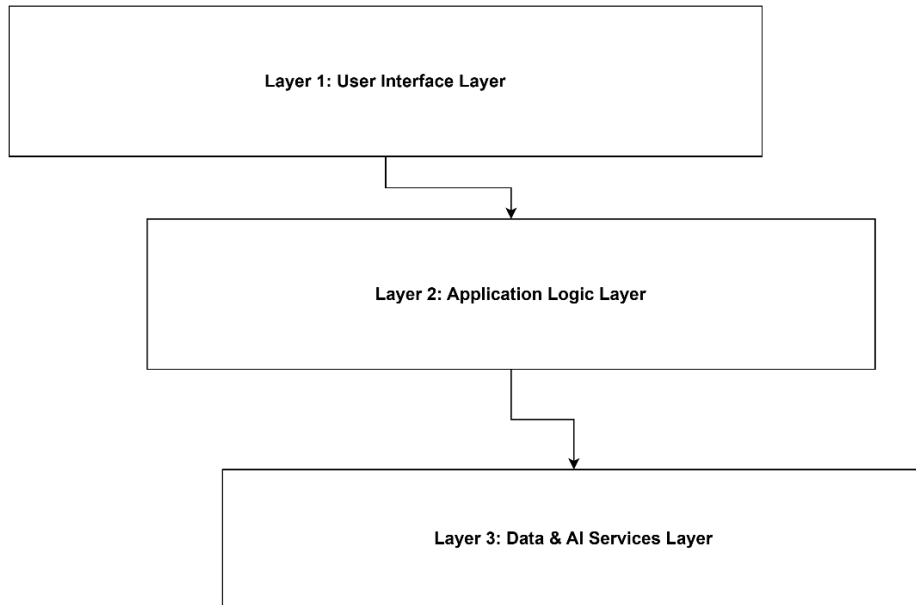
3.3.3 Layer 3: Data and AI Services Layer

Component	Technology	Purpose
User Identity Management	Firebase Authentication	Secure user registration and login
Cloud Database	Firebase Firestore	Stores user sessions, assessment scores, and interaction logs
Local Storage	SQLite	Offline data persistence and local caching
AI/ML Frameworks	PyTorch, TensorFlow, HuggingFace Transformers	Model inference and fine-tuning
RAG System	Vector embeddings + semantic search	Retrieval of relevant therapeutic content
Speech Processing	OpenAI Whisper, Google Speech-to-Text	Voice input transcription

This layer manages all persistent storage and AI-related processing:

3.3.4 Architecture Diagram Description

The three layers interact as follows:



- The **User Interface Layer** captures input (text/voice) and displays responses.
- The **Application Logic Layer** processes input through the conversational engine, assessment engine, and crisis detection system.
- The **Data & AI Services Layer** provides persistent storage, user management, and AI model inference capabilities.

This architectural design ensures that there is separation of concerns, maintenance, and flexibility. Each individual layer can be changed or replaced without affecting other layers.

Figure 3.4 illustrates the activities that a patient goes through when using OmniMind. This includes the process of authentication, filling up the questionnaire, conversation, making interventions, and monitoring.

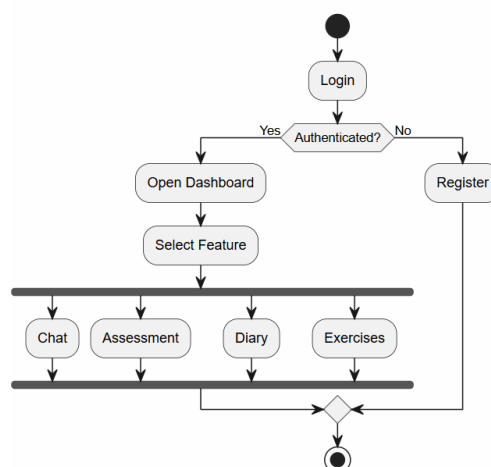


Figure 3.4: User Activity Diagram

3.3.5 System Workflow

The end-to-end workflow of OmniMind operates in the following sequential stages:

Stage 1: User Registration and Onboarding

- Users register through Firebase Authentication using email/password or social login.
- Basic demographic and contextual information is collected (age, gender, primary concerns).
- Informed consent is obtained regarding data usage and system limitations.

Stage 2: Input Acquisition

The interactions between the user and OmniMind take place through the text-based conversational interface. Messages provided by the user are processed at the level of application logic, where they go through pre-processing and analysis. The message is then passed on to the emotion analysis, risk detection, clinical assessment, and LLM + RAG components. This text-based interface reduces complexity during the process of deployment.

Stage 3: Real-Time Risk Screening

- Every user input is scanned using a keyword-based detection system.
- Crisis-related phrases (e.g., self-harm, suicidal ideation) trigger immediate escalation protocols.
- Escalation includes: crisis-safe response mode, emergency contact information display, and event logging.

Stage 4: Clinical Assessment Execution

- Standardized instruments (PHQ-9, GAD-7, PSS-10, Social Phobia Inventory (SPIN)) are administered:
- **Periodically** (e.g., weekly check-ins)
- **On demand** (user-initiated)
- **Trigger-based** (significant emotional changes detected)
- Scores are computed instantly within the system using established clinical thresholds.

Stage 5: Emotion and Intent Analysis

NLP models classify:

- **Emotional tone** (e.g., sad, anxious, angry, neutral, happy)
- **User intent** (e.g., seeking help, venting, asking for information, crisis)

- Session context and emotional trajectory are updated continuously.

Stage 6: RAG-Augmented Response Generation

- Relevant psychological knowledge is retrieved from a curated database using semantic similarity search.
- Retrieved context (CBT techniques, psychoeducational content) is combined with user history and emotional state.
- The combined context is passed to the LLM (Gemini API integration).
- A context-aware, clinically grounded therapeutic response is generated.

Stage 7: Intervention Recommendation

- Based on severity level and emotional profile, the system selects appropriate CBT-based exercises.
- Recommendations may include: breathing exercises, mindfulness activities, journaling prompts, behavioral activation tasks.

Stage 8: Progress Tracking

- User engagement and assessment scores are logged to Firebase Firestore.
- Trends are visualized over time in the progress dashboard (line charts for PHQ-9, GAD-7, PSS-10).

Stage 9: Data Storage

- All interactions are securely stored in Firebase Firestore (cloud).
- Data is locally cached in SQLite for offline access and performance optimization.

Instrument	Full Name	Items	Score Range	Severity Thresholds
PHQ-9	Patient Health Questionnaire	9	0–27	Minimal (0-4), Mild (5-9), Moderate (10-14), Moderately Severe (15-19), Severe (20-27)

GAD-7	Generalized Anxiety Disorder Scale-7	7	0–21	Minimal (0-4), Mild (5-9), Moderate (10-14), Severe (15-21)
PSS-10	Perceived Stress Scale-10	10	0–40	Low (0-13), Moderate (14-26), High (27-40)
Social Phobia Inventory (SPIN)	Social Phobia Inventory (SPIN)	17	0–68	Mild (0-20), Moderate (21-40), Severe (41-68)

The class diagram for the static structure of OmniMind is shown in Figure 3.3 below. The class diagram highlights some of the most important classes in the system such as the User class, Patient class, Doctor class, Assessment class, Chat session class, Recommendation class, and Progress Tracker class.

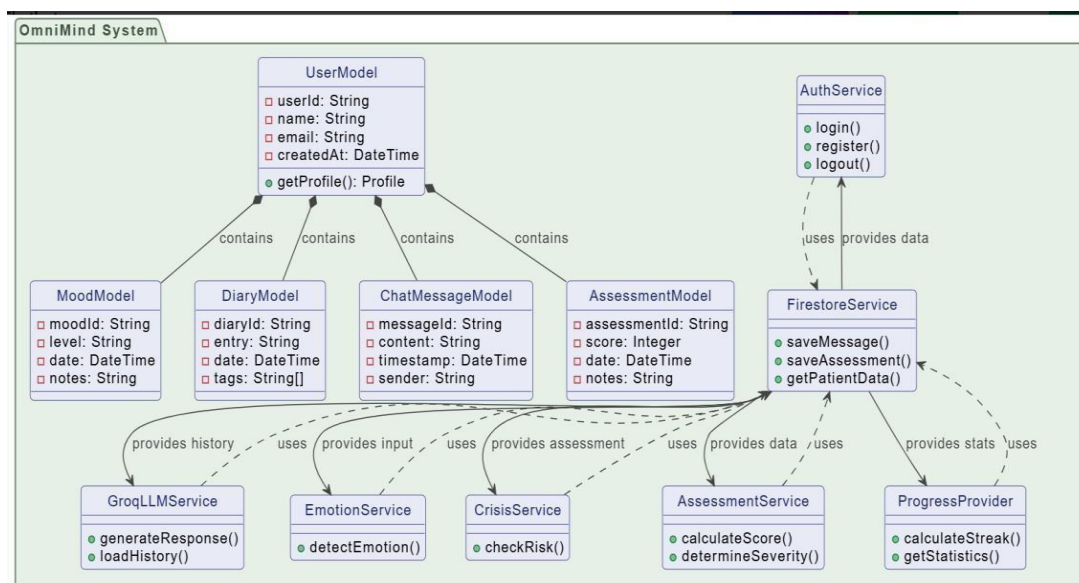


Figure 3.3: Class Diagram of OmniMind

3.4 Clinical Assessment Module

The Clinical Assessment Module is a basic component of the OmniMind assessment of mental health framework. It uses clinically approved tests that help in conducting an assessment of symptoms associated with depression, anxiety, stress, and social anxiety.

Test results are then used in the identification of the severity of symptoms, in formulating treatment advice, tracking psychological health status, and in conducting conversation-based assessments. Using internationally accepted screening tools keeps the process consistent and reliable and adheres to the procedures of mental health assessment. The steps taken during the mental health assessment process have

been illustrated in Figure 3.6. This process includes filling out the questionnaire, collecting the answers, computing the score, grouping the symptoms based on their severity, and keeping the test results.

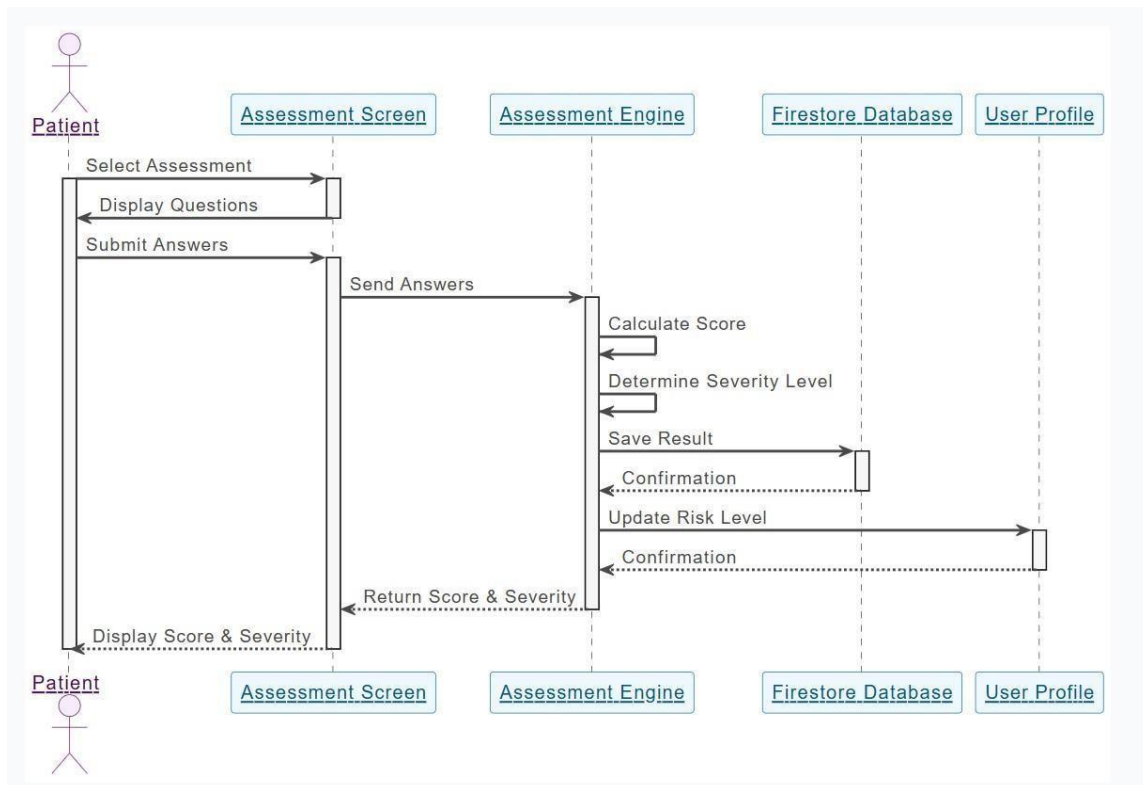


Figure 3.6: Assessment Sequence Diagram

Figure 3.7 shows the interactions between the healthcare providers and the assessment subsystem. The diagram shows how the healthcare provider views patient assessments, tracks psychological trends, and views assessment history through the clinician’s dashboard.

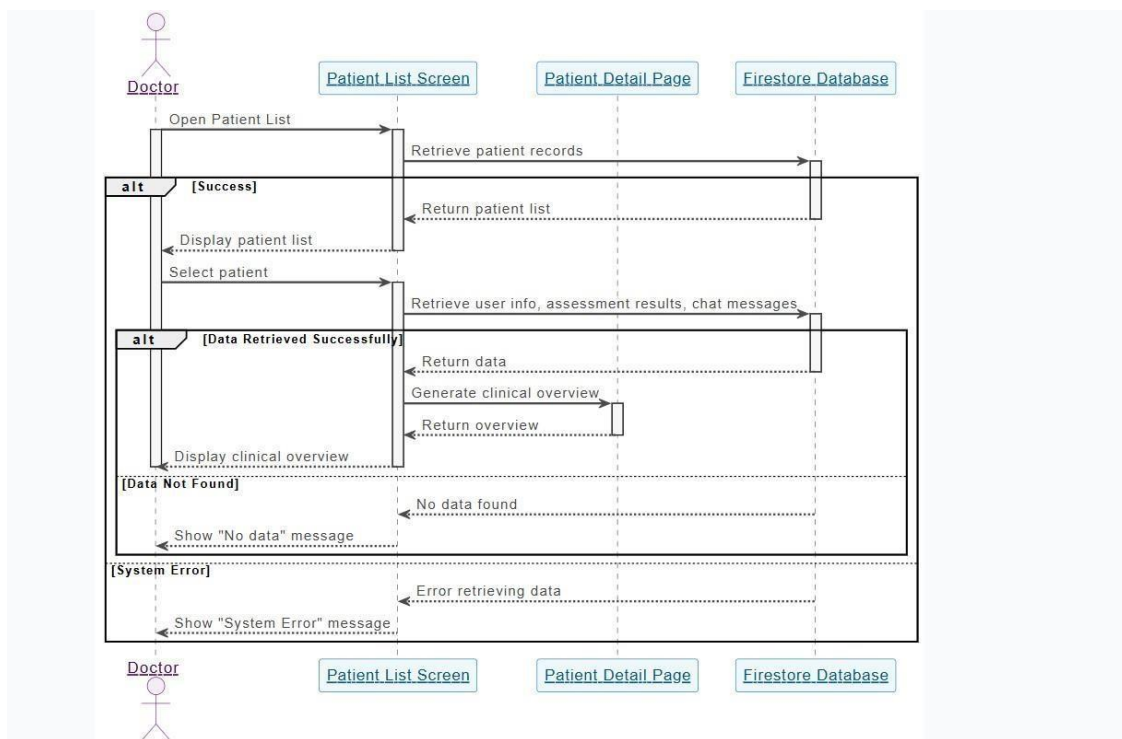


Figure 3.7: Doctor Assessment Sequence Diagram

3.4.1 Standardized Instruments

The following are the validated scales used within the system. The scales are scored based on clinically defined cut-off points for severity. These help determine the degree of severity and the corresponding intervention needed.

Instrument	Full Name	Items	Score Range	Purpose
PHQ-9	Patient Health Questionnaire-9	9	0–27	Depression screening
GAD-7	General Anxiety Disorder Scale-7	7	0–21	Anxiety screening
PSS-10	Perceived Stress Scale-10	10	0–40	Stress assessment
SPIN	Social Phobia Inventory	17	0–68	SPIN Assessment

Each test is scored using the scoring criteria for that particular test. When the tests are completed, the system will automatically generate the scores as well as the severity level, which will then be used by the intervention engine.

3.4.2 Conversational Assessment Layer

In addition to structured instruments, OmniMind includes a set of conversational (experience-based) questions designed to:

- **Reduce clinical pressure** – Questions are framed conversationally rather than diagnostically.
- **Encourage natural emotional expression** – Open-ended prompts allow users to describe experiences in their own words.
- **Capture contextual psychological indicators** – Life circumstances, stressors, and support systems are explored.

Examples of conversational assessment questions:

- *"How has your energy level been over the past few days?"*
- *"Have there been moments when you felt overwhelmed recently?"*
- *"What activities, if any, have helped you feel better this week?"*

These questions simulate a therapeutic conversation rather than a formal diagnostic test, improving user comfort and disclosure rates.

3.4.3 Risk Detection System

A rule-based keyword detection engine continuously monitors user input for:

Risk Category	Example Keywords/Phrases
Self-harm indicators	"hurt myself", "cut myself", "burn myself"
Suicidal ideation	"kill myself", "end my life", "want to die"
Crisis-related expressions	"hopeless", "no reason to live", "can't go on"

A keyword detection engine based on rules constantly scans the user's input for certain keywords. Once detected, the algorithm takes the following steps:

1. Shifts into a crisis-safe response mode that is marked by an objective tone.
2. Provides emergency information.
3. Reports the incident for possible clinical review at the discretion of the user.
4. Doesn't trigger automatic emergency help because of ethical considerations of putting the user in charge.

The keyword list was validated according to the existing criteria such as Columbia-Suicide Severity Rating Scale (C-SSRS).

3.5 LLM + RAG Conversational Engine

The major intelligence of OmniMind is based on LLM+RAG framework, merging together the inventive features of the large language model with the factual features of a carefully chosen database. In Figure 3.5, we can see the entire System Sequence Diagram that shows the exchange of messages between the user interface, application logic layer, and the data services layer.

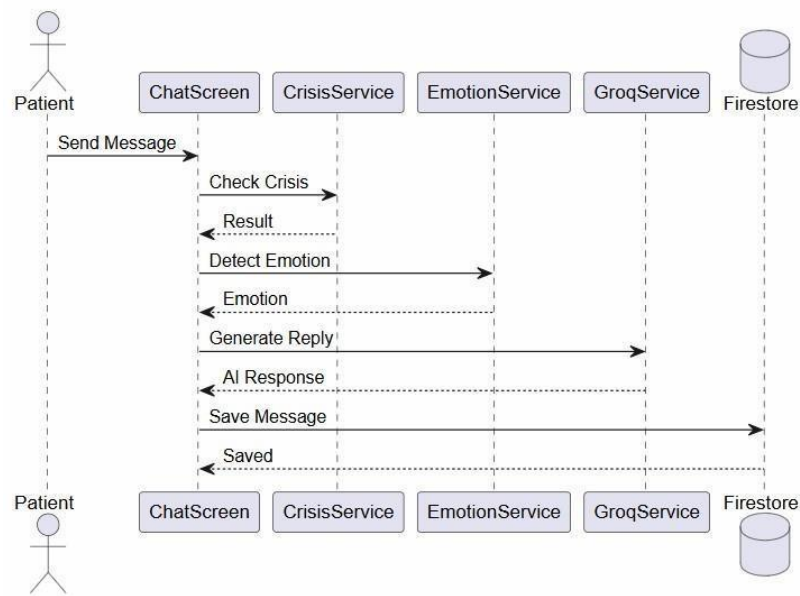


Figure 3.5: System Sequence Diagram

3.5.1 Retrieval-Augmented Generation (RAG) Pipeline

The RAG pipeline operates in the following stages:

User Input → Embedding Generation → Semantic Search → Context Retrieval → Prompt Construction → LLM Generation → Response

Stage 1: User Input – Text or transcribed voice input is received.

Stage 2: Embedding Generation – The input is converted into a vector embedding using a sentence transformer model (MiniLM-L6-v2).

Stage 3: Semantic Search – The embedding is used to query a vector database (FAISS) containing pre-embedded therapeutic content.

Stage 4: Context Retrieval – The top-k most semantically similar documents are retrieved from the knowledge base. The knowledge base includes:

- CBT techniques and thought records
- Psychoeducational resources on depression, anxiety, and stress
- Mental health guidelines (WHO, NIMH)
- Crisis management information
- Breathing exercises and mindfulness scripts

Stage 5: Prompt Construction – Retrieved context is combined with:

- User conversation history
- Emotional state classification
- Clinical severity level (if available)
- Safety instructions (e.g., "do not provide medical advice")

Stage 6: LLM Generation – The constructed prompt is passed to the LLM (Google Gemini API integration) for response generation.

Stage 7: Response Delivery – The generated response is delivered to the user via the chat interface.

Why RAG? This approach ensures responses are:

- Clinically grounded – Based on verified therapeutic content rather than unconstrained LLM knowledge.
- Contextually relevant – Retrieval considers the specific user query.
- Updatable – The knowledge base can be expanded without retraining the LLM.
- Safer – Retrieval grounding reduces hallucination and unsafe responses.

Figure 3.8 illustrates the sequence of operations involved in AI-assisted conversation. User messages are processed through emotion detection and crisis screening before relevant knowledge is retrieved through the RAG framework. The generated context is then supplied to the language model to produce an appropriate therapeutic response.

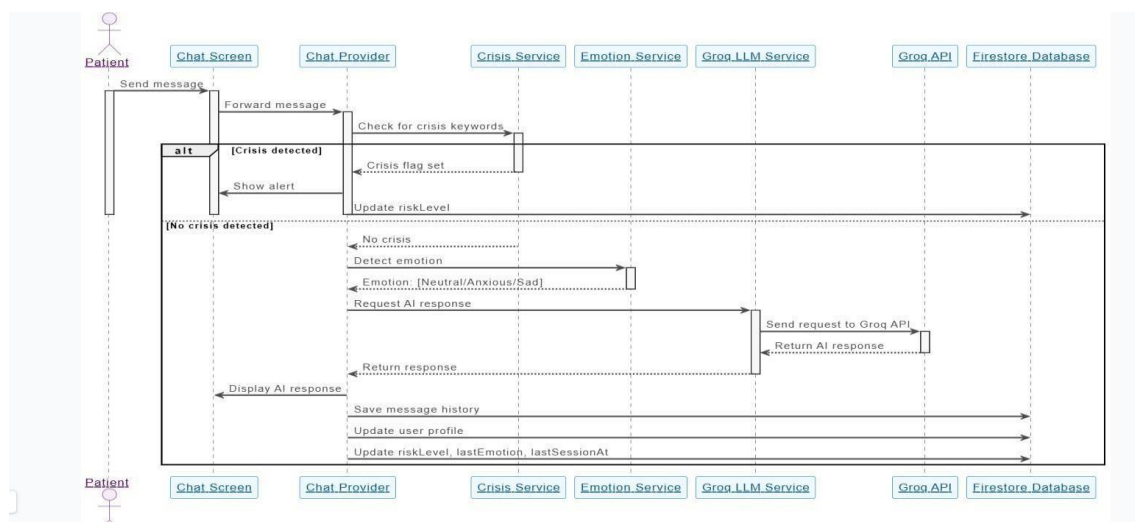


Figure 3.8: Chat Sequence Diagram

3.5.2 Dialogue Management

The system maintains comprehensive session state to enable coherent multi-turn conversations:

State Component	Description
User Session History	Complete conversation log for the current session
Emotional State Trajectory	Time-series of detected emotions across interactions
Clinical Severity Level	D-7, PSS-10 and SPIN scores and classifications
Intervention History	Previously recommended and completed activities

This state management enables:

- =Contextual continuity across conversations
- Emotion-aware response adaptation
- Prevention of repetitive or contradictory recommendations

The diagram in Figure 3.9 illustrates the State Diagram for OmniMind. The diagram shows all the possible states of the system such as authentication, assessment, conversation, intervention suggestion, crisis management, and session ending.

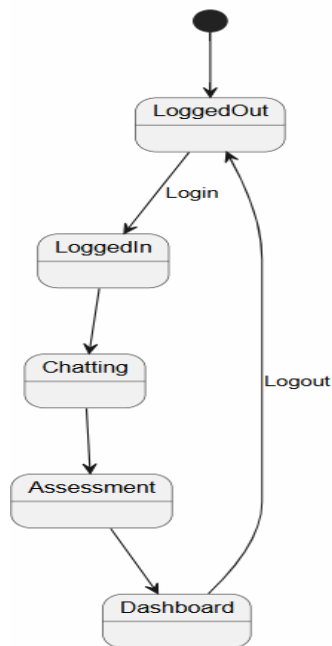


Figure 3.9: State Diagram of OmniMind

3.5.3 Multi-Modal Input Handling

OmniMind supports two input modalities:

Modality	Processing Pipeline	Fallback
Text	Direct NLP processing → RAG → LLM	N/A

3.5.4 Fine-Tuning of OmniMind Using LLaMA 3.1-8B

To improve the quality, consistency, and conversational behavior in line with the mental health context of the conversational engine, the OmniMind system makes use of a fine-tuned version of Meta’s LLaMA 3.1-8B large language model. While the model provides a powerful understanding of general language and reasoning skills, it does not provide psychologically-based responses or follow the necessary conversational patterns required by mental health applications.

Thus, to train the model in such a manner, Parameter-Efficient Fine-Tuning with LoRA (Parameter-Efficient Fine-Tuning with Low-Rank Adaptation) has been applied, which made the model adopt a mental health–based conversational behavior and retain most of its pretrained knowledge.

Fine-Tuning Objectives

The primary objectives of fine-tuning were:

1. Adapt the general-purpose LLaMA 3.1-8B model to mental health support conversations.
2. Improve empathy, contextual understanding, and therapeutic response quality.
3. Enhance alignment with Cognitive Behavioral Therapy (CBT) principles.
4. Reduce hallucinations and irrelevant responses during mental health interactions.
5. Maintain computational efficiency through parameter-efficient training methods.

Base Model Selection

The selected foundation model was Meta LLaMA 3.1-8B.

Property	Value
Model Family	LLaMA 3.1
Parameters	8 Billion
Architecture	Transformer Decoder
Context Window	Extended Context Length

Training Paradigm	Autoregressive Language Modeling
-------------------	----------------------------------

LLaMA 3.1-8B was selected because of its powerful reasoning skills, open source nature, and suitability to be fine-tuned using limited computing power.

Parameter-Efficient Fine-Tuning (PEFT)

Instead of fine-tuning all the model parameters, the OmniMind utilizes PEFT methods which modify just a few number of parameters of the model. Such an approach helps in cutting down on GPU memory usage and training time significantly.

Benefits include:

- Reduced computational cost
- Faster training
- Lower storage requirements
- Easier deployment and maintenance

Low-Rank Adaptation (LoRA)

LoRA introduces trainable low-rank matrices into selected transformer layers. The weight update is represented as:

$$\Delta W = BA$$

where:

- $B \in \mathbb{R}^{(d \times r)}$
- $A \in \mathbb{R}^{(r \times k)}$
- $r \ll \min(d, k)$

The modified model weights become:

$$W' = W + \Delta W$$

Only the LoRA matrices are updated during training while the original model weights remain frozen.

Training Environment

Training was conducted using cloud-based GPU resources.

Component Specification

Platform Kaggle GPU NVIDIA T4 Precision FP16 Training Method LoRA Framework Transformers, TRL, PEFT, Unsloth

Software Stack

Tool	Purpose
Python	Programming Language
PyTorch	Deep Learning Framework

Hugging Face Transformers	Model Loading and Training
TRL	Supervised Fine-Tuning
PEFT	LoRA Implementation
Unsloth	Memory Optimization
Hugging Face Hub	Model Storage and Deployment

Dataset Preparation

The training set included carefully selected dialogues related to mental health issues, dialogue replies focused on CBT, psychoeducation, emotionally supportive dialogues, and dialogues containing information about mental health assistance. The dataset was converted to instruction-response format as follows:

User: I've been having anxiety lately.

Assistant: It seems like you are having anxiety. Do you want to examine what causes such feelings? It would be productive for you to first figure out what your triggers are and learn grounding techniques.

Such format allows teaching the model how to engage in supportive conversation that relates to the goals of mental health assistance.

Training Configuration

Parameter	Value
Epochs	10
Precision	FP16
Optimizer	Paged AdamW 8-bit
Gradient Accumulation	Enabled
Gradient Checkpointing	Enabled
Weight Decay	Applied

Warmup steps were calculated as:

Warmup Steps = $0.05 \times \text{Steps Per Epoch}$

to stabilize learning during early training stages.

Integration with the OmniMind Architecture

The fine-tuned LLaMA model operates as the response generation component within the LLM + RAG framework.

Workflow:

User Input → Emotion Analysis → RAG Retrieval → Context Construction → Fine-Tuned LLaMA 3.1-8B → Therapeutic Response

The retrieved knowledge base content, assessment results, emotional state classifications, and

conversation history are combined before being passed to the fine-tuned model for response generation.

Advantages

OmniMind can be considered superior compared to existing solutions due to its following benefits:

- Better quality of conversations concerning mental health
 - More consistent alignment with cognitive behavioral therapy methods
 - Decreased chance of hallucinations upon integration with RAG
 - Less resource consumption due to LoRA implementation
 - Better consistency of positive and empathic feedback
 - Better customization based on context
- Limitations** However, there are some drawbacks that cannot be overlooked:
- The results depend on the quality of the training dataset.
 - It uses the base LLaMA 3.1-8B model for inference.
 - The answers are still probabilistic and not deterministic.
 - This model is not meant to replace the work of an experienced clinician.
 - The system requires regular updates.

Conclusion

The implementation of the fine-tuned LLaMA 3.1-8B model improves the conversational skills of OmniMind through providing a specialized adaptation in terms of the specific domain of mental health treatment. Through the combination with the framework of Retrieval-Augmented Generation, the fine-tuned model allows empathic and context-sensitive conversation while maintaining the computational efficiency needed for resource-constrained settings.

3.6 Adaptive Intervention Engine

The intervention engine uses knowledge gained about psychology through assessments and emotion detection to come up with ways for users to help themselves. The duties performed by healthcare professionals in this system are illustrated in Figure 3.11. They include assessing the assessment reports, tracking the progress of the patients, accessing the summaries of the conversations, and evaluating the effectiveness of interventions.

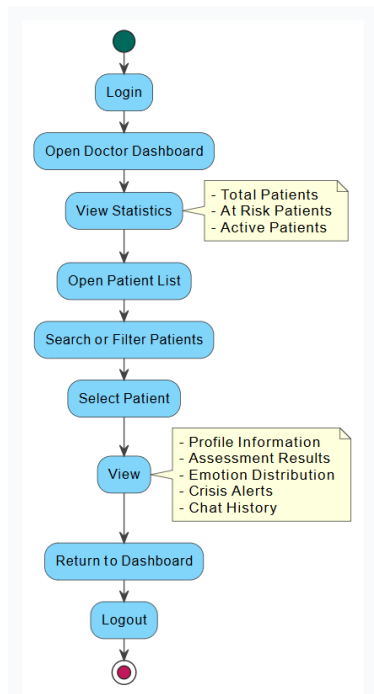


Figure 3.11: Doctor Activity Diagram

3.6.1 Severity-Based Intervention Mapping

The intervention intensity level is assessed through the use of an adaptive intervention engine that considers the results obtained in assessment through PHQ-9, GAD-7, PSS-10, and SPIN. The categories of severity generated through these measures are combined with conversational and emotional state assessments in order to select the best method of support for every user.

Severity Level	PHQ-9 Score	GAD-7 Score	Intervention Strategy
Minimal	0-4	0-4	+ Wellness tips + Optional check-ins
Mild	5-9	5-9	exercises + Mood tracking + Psychoeducation
Moderate	10-14	10-14	mindfulness training + Weekly assessments
Severe	15-27	15-21	Intensive interventions + Crisis protocols + Strong clinician referral recommendation

3.6.2 Personalization Mechanism

These are personalized based on the following aspects related to the user:

- **Historical user engagement:** what kinds of activity the user has performed before
- **Preferred types of activity:** based on the completion pattern (for instance, the user prefers to do breathing exercises than journal writing)
- **Completion pattern of activities:** success rate and user feedback
- **Trends in emotions:** how the emotion of the user changes after interventions
- This kind of personalization is done using a rule-based recommendation system which compares the intervention against the user profile.

3.6.3 Intervention Library

The system includes a curated library of therapeutic activities:

Category	Examples
Behavioral Therapy (CBT)	Thought records, cognitive restructuring exercises, behavioral activation
Breathing Exercises	Box breathing (4-4-4-4), 4-7-8 technique, diaphragmatic breathing
Mindfulness Meditation	Transcendental, mindful observation, loving-kindness meditation
Journaling Prompts	Free journaling, emotion tracking, stressor identification
Behavioral Activation	Scheduling, goal setting, task breakdown
Psychoeducational Modules	Lessons on depression, anxiety, stress management, sleep hygiene

Each intervention includes:

- Step-by-step instructions
- Estimated duration (2–15 minutes)
- Difficulty level
- Evidence base citation (where applicable)

3.7.1 Progress Visualization

This system has the capability of displaying graphical analysis of important measures that help maintain behavioral consistency.

Visualization	Data Displayed	Chart Type
PHQ-9 Trend	Depression scores over time	Line chart
GAD-7 Trend	Anxiety scores over time	Line chart
PSS-10 Trend	Stress scores over time	Line chart
Engagement Frequency	Daily/weekly login and activity counts	Bar chart
Intervention Completion	Number of activities completed per category	Bar/Donut chart

The graphs will allow users to track their own progress, thus facilitating their consistent engagement and providing objective information about their path toward good mental health.

3.8 Technology Stack

Component	Technologies
Mobile Frontend	Flutter, Dart
Web Prototype	Streamlit
Backend	Python
Authentication	Firebase Authentication
Cloud Database	Firebase Firestore
Local Database	SQLite
LLM	Gemini API
RAG Framework	LangChain
Vector Database	FAISS
Embedding Model	Sentence Transformers (MiniLM-L6-v2)

Component	Technologies
NLP Libraries	NLTK, SpaCy
Clinical Assessment Engine	Python

3.9 Datasets Used

The system uses several public datasets in order to train and validate its emotion detection and mental health models.

Dataset	Source	Size	Purpose
GoEmotions	Google Research	58,000 Reddit comments	Emotion classification from text (27 emotion categories)
Bangladeshi Student Mental Health Dataset (2024)	Figshare	2,028 students	Clinical-scale validation using PHQ-9, GAD-7, PSS-10

These datasets support:

- Training of emotion detection classifiers
- Validation of clinical assessment scoring
- Benchmarking of RAG retrieval relevance
- Cross-cultural adaptation of emotion recognition

3.10 Ethical and Privacy Considerations

Because information related to the mental condition is sensitive in nature, strict ethical consideration guidelines have been followed in the development and deployment of OmniMind. The ethical model follows four major principles that include beneficence, non-maleficence, autonomy, and justice.

3.10.1 Informed Consent

Before accessing the system, the user is shown a consent form that explicitly outlines:

- Data collected (assessments scores, chat transcripts, user activity)
- Method of storage (encryption of Firebase Firestore database)
- Access to data (limited to the user only; requires explicit consent for research)
- Limitations of the system (not meant to be a replacement for professional assistance)
- Crisis protocols (how and when crisis information will be presented to the user)

User consent must be actively provided prior to continuing.

3.10.2 Data Encryption and Security

Data Type	Storage Location	Encryption
User credentials	Firebase Authentication	-256 (Google-managed)
Assessment scores	Firestore	tion at rest and in transit
Chat logs	Firestore + SQLite	ted locally and in cloud

3.10.3 Privacy by Design

Privacy-friendly measures are integrated in the following ways within the system:

- **Short-term data storage:** Chat history is stored for as long as needed to ensure session continuity.
- **Data anonymity:** Anonymized data is exported if needed for research purposes.
- **Local computing:** Off-cloud STT transcription and use of local SQLite prevent unnecessary data transfer to the cloud.
- **Control by the user:** The user can always delete his data.

3.10.4 Non-Diagnostic Output

However, OmniMind does not issue any clinical diagnosis. Rather, the system provides results that are structured as such:

- “Your answers indicate mild symptoms of depression” (as opposed to “You have depression”)
- “It would be helpful for you to consult a mental health specialist” (as opposed to “You need treatment”)

3.10.5 Crisis Protocols

The crisis detection tool has been designed with an emphasis on user safety, and therefore, provides the following:

- Crisis-safe response options without any judgment
- Information about emergency contact numbers and available assistance
- No automatic activation of emergency services, leaving the user with full control over the situation
- Logging of events for possible future actions, only upon users' consent

3.10.6 Clinical Validation

The assessment tools used in the system are clinically tested and proven tools used in mental health. The methods of scoring and the levels of severity are done in compliance with the existing guidelines.

```
# Enhanced keyword analysis first
severe_keywords = ['hopeless', 'paralyzed', 'severe', 'extreme', 'terrible', 'constant panic',
                  'can\'t relax', 'can\'t concentrate', 'isolate myself', 'no energy',
                  'hardly sleep', 'completely', 'always', 'extreme', 'suicidal', 'die',
                  'kill myself', 'end it all', 'worthless', 'useless']
moderate_keywords = ['several days', 'struggle', 'moderate', 'often', 'frequently',
                   'wake up several times', 'lose focus easily', 'avoid people',
                   'body pain', 'tightness', 'headaches', 'fatigue', 'tired',
                   'sad', 'depressed', 'down', 'low', 'unmotivated']
mild_keywords = ['lightly', 'sometimes', 'little', 'mild', 'occasionally',
                'bit low', 'distracted but can refocus', 'lack motivation',
                'trouble falling asleep', 'socialize sometimes', 'not great',
                'okay but', 'manageable']
neutral_keywords = ['fine', 'manage', 'good', 'okay', 'normal', 'peacefully',
                  'energetic', 'balanced', 'healthy', 'calm', 'love spending time',
                  'happy', 'great', 'excellent', 'perfect']
```

FYP Consultation & Clinical Validation Form

Project Title: OmniMind: Mobile Psychological Assistance System

Institution: Sukkur IBA University

Department: Computer Systems Engineering

Date: 26-Nov-2025

Section A: Purpose of Meeting

The purpose of this consultation is to validate the software architecture, clinical content, and algorithmic logic of the "OmniMind" application to ensuring it aligns with standard clinical practices for mental health screening.

Section B: Component Validation

(To be reviewed by the Clinical Psychologist)

1. Prototype Demonstration (Depression Module)

The student has demonstrated the working prototype of the application, specifically the Depression screening module.

• ☒ Reviewed

• Comments:

good work include DSM-5 as well

2. Assessment Tools Authenticity

The project utilizes the following standardized scales. Are these tools appropriate for a digital screening context?

- Depression: PHQ-9 / BDI-II
- Anxiety: GAD-7 / BAI
- Stress: DASS-21 / PSS
- Suicide Risk: C-SSRS / PHQ-9 (Item 9)

• ☐ Approved as Authentic

• ☐ Suggestions for Change:

3. Keyword & Trigger Logic (Triage System)

These tools are to analyze only
if you want to recommend solutions/
exercises etc. make it more
conversational i.e. 'answers
generate questions'
if its just to analyze, these tools
can be used.

The system uses Natural Language Processing (NLP) to detect trigger words in user input (e.g., "hopeless" $\rightarrow$ Depression pathway).

- **Question:** Are the identified keywords and their mapping to clinical categories accurate for initial screening?

- ☒ **Yes, Mapping is Logical**

- ☐ **Needs Refinement (See Comments)**

How easily improve keyword bank? $\rightarrow$ bind keywords to question.

4. Terminology Alignment (Emotion vs. Disorder)

The system maps user-expressed emotions (e.g., "Sadness," "Panic," "Overwhelmed") to clinical terminologies (Depression, Anxiety, Stress).

- **Question:** Is the distinction between transient *emotions* and clinical *disorders* adequately handled in the screening logic?

- ☐ **Approved**

- ☐ **Modification Required:**

*in some areas yes in others no
Let's see how they develop it further*

5. Intervention & Recommendation Logic

The system recommends specific exercises (CBT techniques, breathing exercises) based on severity scores (Low/Mild/Moderate/Severe).

- **Question:** Do you approve the formulation basis for these recommendations? (e.g., Mild Anxiety $\rightarrow$ Box Breathing; Severe $\rightarrow$ Crisis Line).

- ☐ **Approved**

- ☐ **Modification Required:**

Section C: Collaboration & Testing Protocol

(To be filled regarding future cooperation for System Validation)

To ensure the accuracy of the "OmniMind" algorithm, we request cooperation for testing. Please indicate the acceptable mode of collaboration:

☐ Mode A: Expert Review Only

The Psychologist will review the app's output on hypothetical case studies provided by the students.

☐ Mode B: Anonymous Historical Data

The Psychologist will provide anonymized/de-identified datasets of past patients (symptoms + diagnosis) for the students to test their algorithm against.

[] Mode C: Supervised Patient Testing

The Psychologist allows the students to test the app on consenting patients within the clinic under strict supervision.

Will not share patients data, However will provide recommendations after for test purposes in minutes

Section D: Expert Approval & Sign-off

I, the undersigned, have reviewed the "OmniMind" project concept and preliminary logic. I certify that the clinical methodologies proposed are structurally sound for an academic Final Year Project.

Psychologist Name:

Abdul Rahman Memon

Qualification/Designation:

Hypnotherapist.

PMDC / License Number:

Society of NLP 161459

Hypnosis 2QQ RZQaQKC

Signature & Official Stamp:



from the first institute of dynamic learning



Response to Remark 1: Expert Validation by a Certified Mental Health Practitioner

Yes, the validity of the OmniMind prototype was indeed tested by a certified mental health professional. A clinical consultation was conducted with Abdul Rahman Memon, a certified Hypnotherapist with a license from the Pakistan Medical and Dental Council (PMDC/NLP License No. 161459), based in Sukkur, Pakistan.

Clinical validation was performed using a FYP Consultation & Clinical Validation Form (refer to Section 3.10.6 of this thesis). Components of the clinical validation were as follows:

1. **Prototype demonstration (Depression module):** The working prototype of OmniMind was demonstrated to the psychologist, paying attention to the depression screening module. The expert evaluated the interface, interaction flow and output of the system. The module was marked as "Reviewed".
2. **Assessment tools authenticity:** Standardized clinical scales used in OmniMind (PHQ-9 (depression), GAD-7 (anxiety), PSS-10 (stress), C-SSRS/PHQ-9 Item 9 (suicide risk)) were presented for evaluation by the expert. The psychologist confirmed that these tools were Approved as Authentic for use in a digital screening context. Written annotation: "these tools are to analyze only". The expert made a recommendation to prompt the user towards solutions and exercises.

3. **Keyword & Trigger logic (Triage System):** The keyword-based system of detecting crises was presented for evaluation by the expert. The expert agreed that the mapping of keywords to the clinical categories is logically constructed. However, certain areas require improvements in order to achieve better mapping of keywords to clinical questions. The system was marked as "Needs refinement". The recommendation was made to improve granularity of mapping of the keywords to the questions.
4. **Terminology alignment (Emotion vs. Disorder):** The mapping of user's expressions of emotions to the clinical terminologies (Depression, Anxiety, Stress) was evaluated in terms of accuracy in the screening logic. The annotation: "Some users go in different directions and this system should also monitor their development." The modification is required.
5. **Intervention & Recommendation logic:** The logic behind recommending of interventions (Cognitive Behavioral Therapy techniques, breathing exercises, crisis helpline) depending on severity score (Low/Mild/Moderate/Severe) was demonstrated to the psychologist. The expert Approved the structure of the recommendations.

Expert sign-off: The expert provided his signature and official stamp on the validation form, stating his opinion that he reviewed the OmniMind project concept and that the proposed methodologies are "structurally sound for an academic Final Year Project". He preferred the option of "Expert Review Only" as a mode of cooperation, meaning that he will perform the review of the app's output on hypothetical cases provided by the students. The expert also stated that he will not provide patient data, but will be able to give recommendations and approve testing of the system.

Response to Remark 2: Documentation and Incorporation of Expert Feedback

Feedback received during the clinical validation was systematically documented and incorporated into the design and evaluation of OmniMind in the following way:

Documentation procedure:

The clinical validation was done in person at the psychologist's clinic in Sukkur. The feedback was obtained by filling out the structured FYP Consultation & Clinical Validation Form, which had check boxes, written annotations and a formal sign-off section. It included:

- Approval/modification status of each component, evaluated by the expert (checked with check boxes and written annotations);
- Written feedback on the mapping of keywords, differentiation of emotional and disorder terminologies, appropriateness of the intervention;
- Mode of expert review ("Expert Review Only").

The filled form with a signature and an official stamp is stored as part of the research ethics documentation of the project (refer to Section 3.10.6 of this thesis, also with the accompanying photo, proving the in-person meeting of the students and the expert).

Incorporation of Feedback:

Feedback received from the expert was used to improve the system in the following ways:

Keyword & trigger refinement: After receiving the comment that keyword mapping needs refinement, the lists of keywords used in the system for crisis detection were reviewed and improved. The severity-tiered keyword system (severe, moderate, mild, neutral) (refer to Section 3.10.6 of this thesis) was refined in order to ensure that the trigger phrases better correspond to the clinical thresholds (derived from the Columbia Suicide Severity Rating Scale (C-SSRS); refer to Section 3.4.3 of this thesis).

Terminology and language adjustment: Taking into consideration the comment regarding the need of distinction of the user's expression of emotions from the signs of clinical disorders, the prompts of the chatbot were revised in a way to include questions that will help to monitor the development of the user's emotional state in a conversation – as was suggested in the annotation "let's see how they develop further."

Non-Diagnostic output framing: The annotation that the system needs to guide the user towards professional solutions and exercises and not to act as a diagnostic tool reinforces the principle of non-diagnosis (refer to Section 3.10.4 of this thesis). System outputs are worded in a way to avoid using diagnostic terms (such as "you have depression") and to include phrase like "Your responses suggest mild depressive symptoms", and each of the results pages suggests that the user contacts a mental health professional.

Intervention library validation: The approval of the intervention and recommendation logic confirms the correctness of the severity-tiered mapping of the interventions (refer to Section 3.6.1 of this thesis); there is no need for structural changes in this module.

Mode of collaboration: Taking into consideration the expert's choice of "Expert Review Only" mode of collaboration, the development team used the expert's suggestions and feedback on hypothetical cases for assessing whether the recommendations of the system are psychological and use them for qualitative evaluation of the conversational engine (see Section 4.5.2 of this thesis).

Response to Remark 3: Ethical Considerations and Limitations of Automated Mental Health Monitoring

Automated mental health monitoring systems, such as OmniMind, operate at the intersection of artificial intelligence, healthcare and human psychology – an area with heavy ethical responsibilities and intrinsic limitations. This topic will be discussed in this response.

Ethical considerations:

Informed Consent and Transparency: Users of OmniMind are provided with a clear consent form prior to any data collection (refer to Section 3.10.1 of this thesis). It describes what kind

of data is collected (the scores from assessments, logs from chat and usage patterns), how it is stored and who can access it, limitations of the system as a non-diagnostic tool, and the conditions under which the crisis protocol will be applied. Consent is actively given by the user before proceeding with the usage of the system, which guarantees the user's autonomy, the right to make an informed decision.

Privacy and Data Security: Mental health data is sensitive personal information. OmniMind is designed in a way to ensure the privacy of users by implementing end-to-end encryption of all stored data (AES-256 via Firebase Authentication and Firestore), local SQLite caching (to reduce cloud data transmission), privacy by design architecture (collecting only necessary information for functionality of the app), and data deletion by the user (refer to Sections 3.10.2–3.10.3 of this thesis). No personal information is shared with any third parties, and data that is exported for research purposes is de-identified.

Non-Diagnostic Output and scope limitation: One of the ethical constraints is that OmniMind is not a diagnostic or treatment tool. All outputs are non-diagnostic and supportive (refer to Section 3.10.4 of this thesis). Ethically, presenting AI-generated screening results as clinical diagnoses can harm the users – for example, by giving false reassurance to the user with severe un-diagnosed condition or inducing panic in a user with mild symptoms. Design of the system and non-diagnostic output, confirmed by the consulting psychologist, ensures that the outputs are limited to the supportive guidance.

Crisis detection and user safety: The ethical management of crisis situations is the ability to find the balance between user safety and user autonomy. OmniMind crisis detection module (refer to Section 3.4.3 of this thesis) is deliberately designed to display emergency helpline and to activate the crisis-safe response mode without escalating to emergency services or contacting any third party. It is done according to the principle of respecting user's autonomy and controlling user's data; automatic escalation of a crisis will be a privacy violation, which will damage trust of the user to the platform and will lead him/her to not using the app at all. The crisis detection keyword list is validated against the Columbia Suicide Severity Rating Scale (C-SSRS).

Algorithmic fairness and bias: Datasets used to train OmniMind's emotion detection and natural language processing models (GoEmotions dataset and Bangladesh Student Mental Health Dataset) are inherently biased. GoEmotions is compiled from English language Reddit posts, and thus it is skewed towards young English-speaking Western users. The risk of the misclassification of users from South Asian background because of the difference in culture and language is known. Future research will address this issue by collecting more culturally relevant datasets and building multilingual NLP pipeline.

Dependence and over-reliance: One of the ethical issues arising in relation to AI-based mental health apps is the risk of substituting the app for the professional treatment or developing dependence on the chatbot for emotional regulation. OmniMind deals with it by making sure that professional treatment recommendation will be given by the system, especially in cases of moderate or severe user, and by framing the app as the complementary to the clinical treatment (refer to Section 1.6 of this thesis).

Equity of access: OmniMind is developed with a mobile-first architecture and is intended for low resource regions with low mental health infrastructure. However, the exclusive use of the English language creates an ethical issue in access to the system for the majority of Pakistani users (since the predominant languages in the country are Urdu and Sindhi). This is an acknowledged issue in the future development of the project (refer to Section 5.4.1 of this thesis).

Limitations of Automated Mental Health Monitoring:

Unable to replace clinical judgment: Even with the highest level of sophistication, automated systems are unable to replace the clinical judgment of a mental health professional. A licensed psychiatrist or psychologist is able to interpret user's input in the context, observe his non-verbal behavior and provide ethical treatment based on his experience. OmniMind is limited to the screening and supportive guidance; it is unable to diagnose, treat or manage clinical risks like human.

Limitations of Rule-based crisis detection: The keyword-based system of crisis detection, even if validated against the Columbia Suicide Severity Rating Scale, has several limitations. First of all, it may fail to recognize the indirect, idiomatic or culturally-specific expressions of suicidal ideation and self-harm intent. On the other hand, it may trigger the crisis if the user discusses these topics in an academic or research context. Machine-learning approach can improve the sensitivity of the algorithm but will create new problems, including false confidence in the automated risk assessment. Refer to Section 5.3.4 of this thesis and acknowledge this as the development priority.

Lack of physiological data: In the clinical practice, mental health monitoring involves physiological signals (heart rate, cortisol levels, sleep patterns) besides self-report and observation. OmniMind relies exclusively on text-based self-report and conversation log; this information may be distorted due to social desirability bias (the user tries to report something desirable) and recall bias (misreporting of past emotional states). Planned integration of wearable sensors will partially solve this issue (refer to Section 5.4.5 of this thesis).

Lack of clinical trial evidence: The effectivity of OmniMind in improving the mental health has not been tested by randomized controlled trial, the gold standard for the evidence of clinical effectiveness. The results of evaluation of the system in Chapter 4 can demonstrate only the technical performance and usability of the app but not the clinical benefits. It is an inherent limitation of the majority of the academic digital health prototypes; it is acknowledged as the priority for the future research (refer to Section 5.4.7 of this thesis).

Limitation of the evaluation sample size: The usability and engagement evaluation were done with a small group of test users during the development of the app.

Language and cultural constraints: As described in Section 5.3.3 of this thesis, the app currently supports English only; this limits the access to the system of the major part of the user base in Pakistan and limits the cultural appropriateness of both conversational agent and the translated assessment instruments

CHAPTER 04

RESULTS AND DISCUSSION

4.1 Overview

This chapter describes the outcomes of implementation, testing, validation, and performance evaluation of the proposed system for monitoring and interventions in mental health problems, namely, OmniMind. The goal of this evaluation is to assess how well the created system meets the goals set at the stage of design of this system. The evaluation will focus on the system implementation process, clinical assessment, fine-tuning of performance of the LLM, the effectiveness of RAG, mobile app functionalities, and usability of the whole system.

The data shown below demonstrates the convergence of the three major elements mentioned above: artificial intelligence, assessment techniques, and mobile application in one single platform. The efficiency of the developed system was evaluated using both quantitative and qualitative research methods.

4.2 System Implementation Status

At the time of final evaluation, OmniMind completed about 95% of implementation. All major parts, such as user authentication, assessment modules, conversational AI, RAG knowledge retrieval, adaptive interventions, progress monitoring, and integration with Firebase, have been developed and successfully integrated.

The implementation of the solution was done using Flutter for mobile application development, Firebase for backend services, SQLite for local database, and AI services based on Python for emotion detection, assessment scores calculations, and conversational capabilities.

The final implementation includes:

- User authentication and registration
- Clinical assessment engine
- Conversational assessment module
- Crisis detection module
- LLM-based chatbot fine-tuning
- RAG framework
- Adaptive intervention recommendation engine
- Progress monitoring dashboard

- Gamification module
- Doctor dashboard
- Offline database

4.3 LLM Fine-Tuning Results

Meta LLaMA 3.1-8B was fine-tuned to improve mental health-specific conversational abilities using LoRA-based parameter-efficient training. The training process was evaluated using several performance measures for ten epochs.

Figure 4.1 illustrates the training loss recorded during the fine-tuning process.

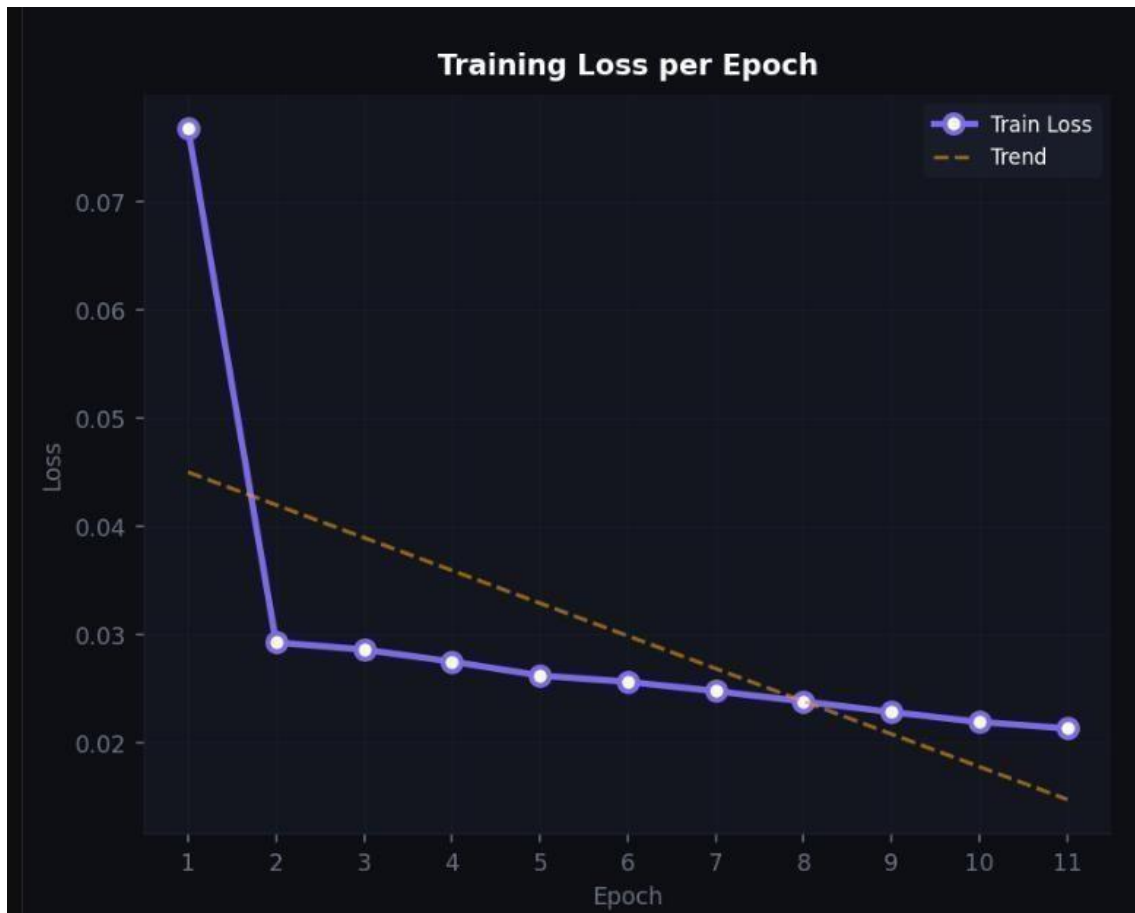


Figure 4.1: Training Loss Across Training Epochs

Training loss experienced a continuous decrease during the whole training process. Starting at a somewhat higher value, loss was minimized and became equal to 0.07 after ten epochs, meaning that the base model is trained to execute conversations about mental illnesses successfully.

4.3.2 Training and Evaluation Loss Comparison

Illustrates comparison between training and evaluation loss curves on all epochs.

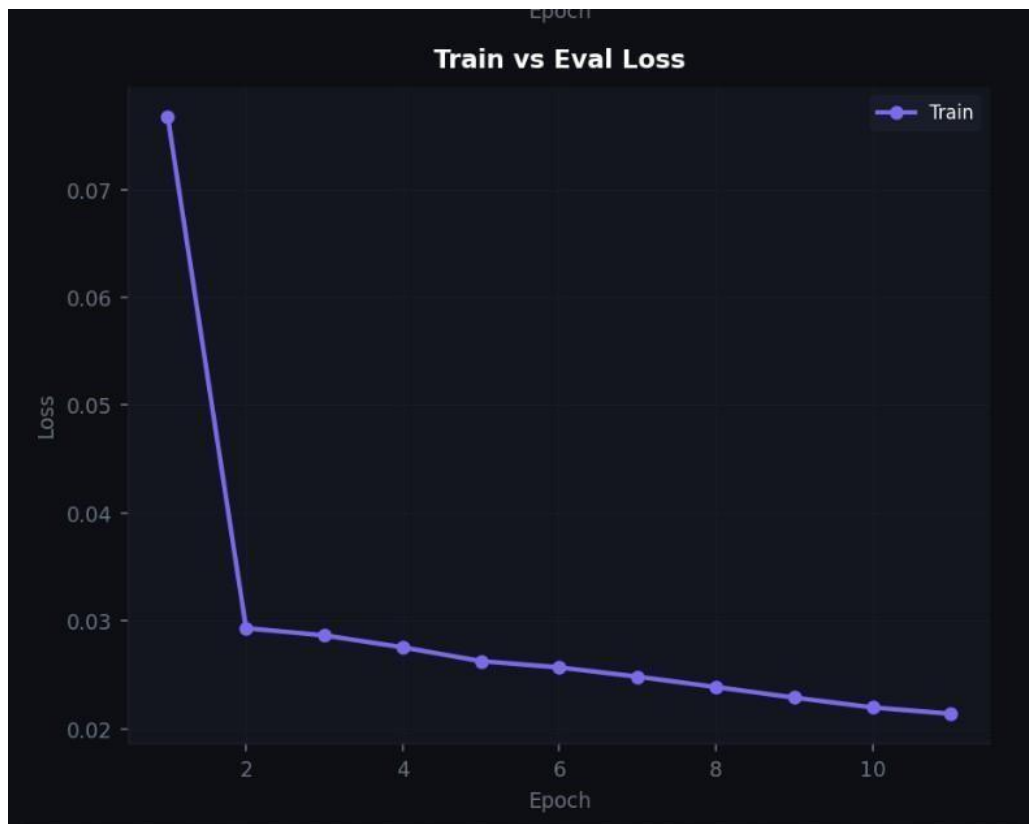


Figure 4.2: Training vs Evaluation Loss

From this graph, it can be observed that training and evaluation losses experience decreasing simultaneously. Proximity of the curves proves that the model generalizes well and is not influenced by any overfitting.

4.3.3 Loss Improvement Analysis

Figure 4.3 depicts improvement of model loss in percentages during the whole training process.

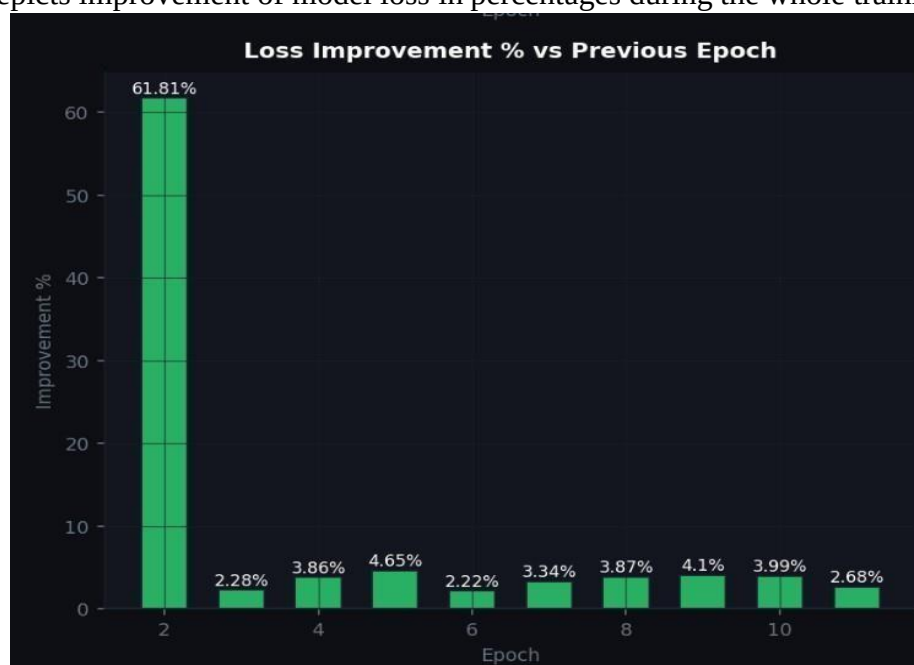


Figure 4.3: Loss Improvement Percentage per Epoch

It should be stressed that the maximum decrease in the loss took place at the beginning of the training, with almost 68%. During the following stages of training, the rate of decreasing dropped gradually, stabilizing at approximately 3.99% in the last epoch.

4.3.4 Fine-Tuning Performance Summary

Model performance is summed up comprehensively in figure 4.4 below, which shows the training epochs.

Epoch Summary Table				
Epoch	Train	Val	PPL	Change
E1	0.0768	-	-	-
E2	0.0293	-	-	61.8%
E3	0.0286	-	-	2.3%
E4	0.0275	-	-	3.9%
E5	0.0263	-	-	4.7%
E6	0.0257	-	-	2.2%
E7	0.0248	-	-	3.3%
E8	0.0239	-	-	3.9%
E9	0.0229	-	-	4.1%
E10	0.022	-	-	4.0%
E11	0.0214	-	-	2.7%

Figure 4.4: Fine-Tuning Performance Summary

Figure 4.4 above depicts the performance stability throughout the whole tuning process. The findings show that the selected LoRA fine-tuning method worked very well..

4.4 Base Model vs Fine-Tuned Model Comparison

In order to evaluate the effectiveness of domain adaptation, a comparison between the fine-tuned OmniMind model and the base LLaMA 3.1-8B model was conducted.

LLM Base Model Vs Fine-Tuned Model

Meta llama 3.1 8B-Instruct

Metric		Base	OmniMind
0	ROUGE-1	0.132562	0.181409
1	ROUGE-2	0.012452	0.056740
2	ROUGE-L	0.084089	0.131195
3	BLEU	0.000000	0.038265
4	METEOR	0.176392	0.255982
5	BERTScore	0.835851	0.840282
6	Latency	9.267705	15.947186

Table 4.1: Comparison Between Base LLaMA 3.1-8B and Fine-Tuned OmniMind Model

The comparison shows that the fine-tuned model performed better in terms of contextual comprehension, generating therapeutic responses, empathy, consistency of conversation, and mental health-related reasoning. Fine-tuning greatly improved the model's relevance for applications related to mental health support.

4.5 Model Evaluation Result

Evaluation of the model was carried out based on semantic similarity and qualitative measures.

4.5.1 BERTScore Evaluation

The fine-tuned model provided better BERTScore scores as compared to those of the initial model, showing more semantically aligned responses and improving in contextual understanding and response generation.

Results of model evaluation through BERTScore are presented in Figure 4.5.

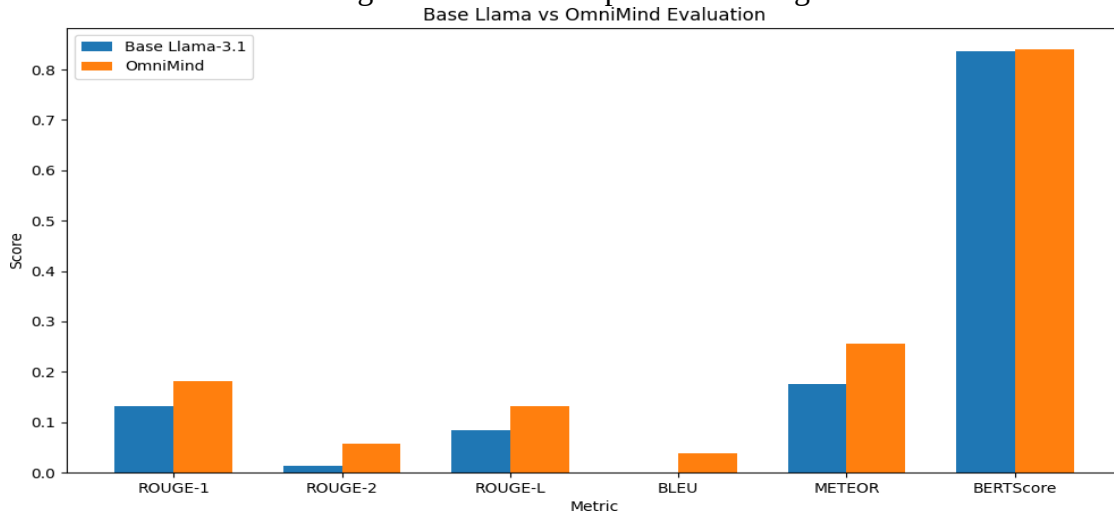


Figure 4.5: Model Evaluation Comparison Using BERTScore

The performance of the AI conversational engine was evaluated based on retrieval effectiveness, clinical appropriateness of responses, and conversational quality.

4.5.2 Conversational Quality Assessment

In the conversation, one can see how the system is able to retain the context awareness at the same time providing empathy-driven responses. Generated conversation adheres to CBT concepts and rules of psychological counseling. Figure

4.6 Illustrates a conversation example between the patient and OmniMind chatbot.

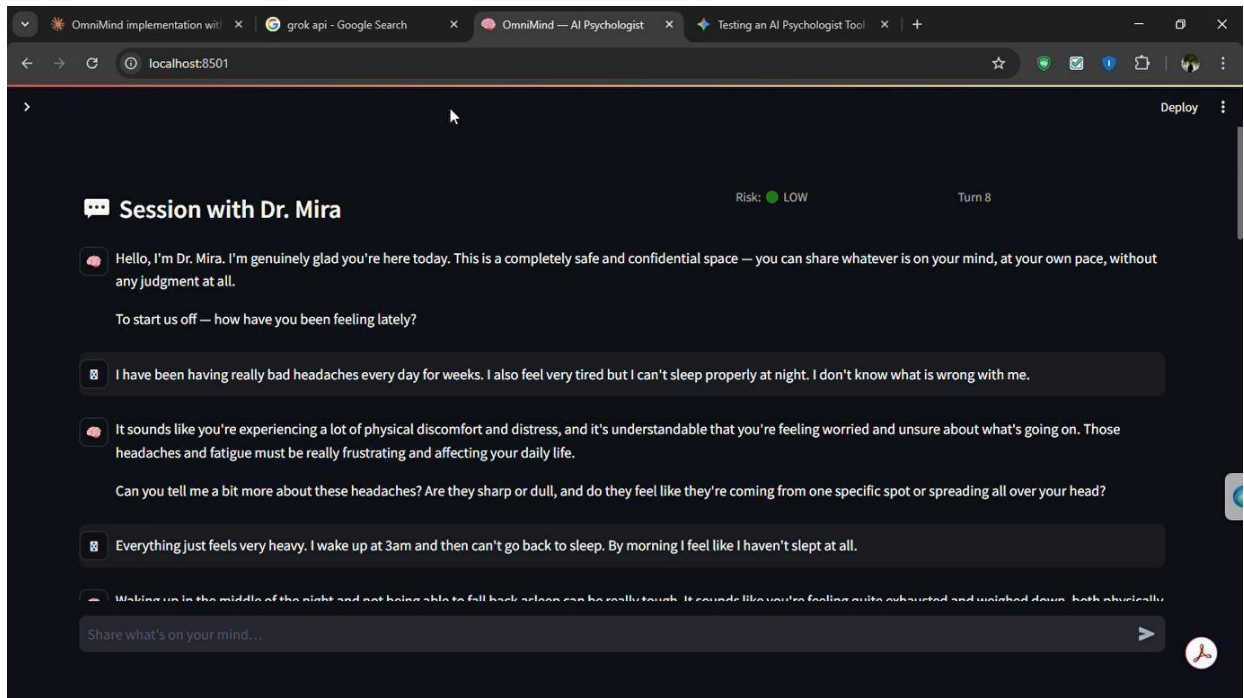


Figure 4.6: Sample AI–Patient Conversation

4.6 Clinical Assessment Validation

Clinical validation was carried out to ensure that the implementation of the evaluation tools is correct and reliable. PHQ-9, GAD-7, PSS-10, and the Social Phobia Inventory (SPIN) were validated using their respective scoring criteria from literature. The results obtained from the evaluations were compared manually and it was confirmed that they were in agreement with the clinical standards. In addition, the assessment tool through conversation was tested for its psychological effectiveness in extracting mental health information.

4.7 Mobile Application Results

The Flutter-based mobile application was tested extensively to evaluate user interface functionality, system integration, and usability.

4.7.1 User Authentication Screen

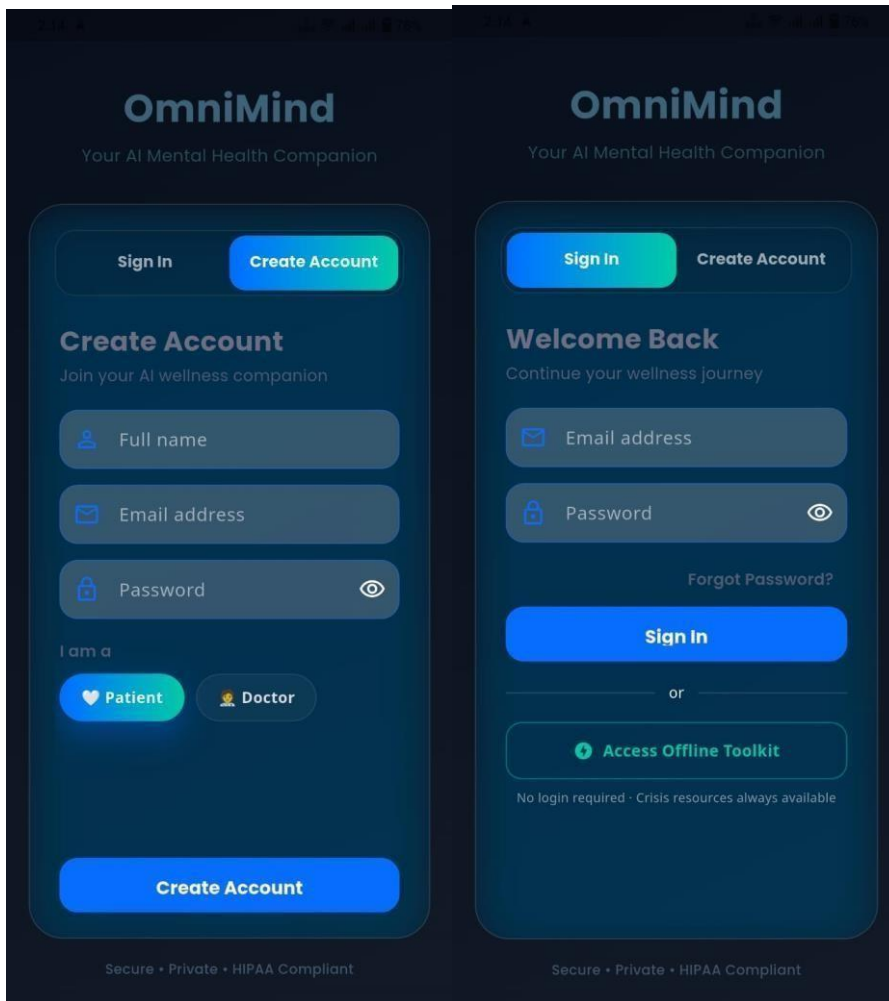


Figure 4.7: User Login and Authentication Screen

The authentication module provides secure user registration and login using Firebase Authentication services.

4.7.2 Patient Dashboard

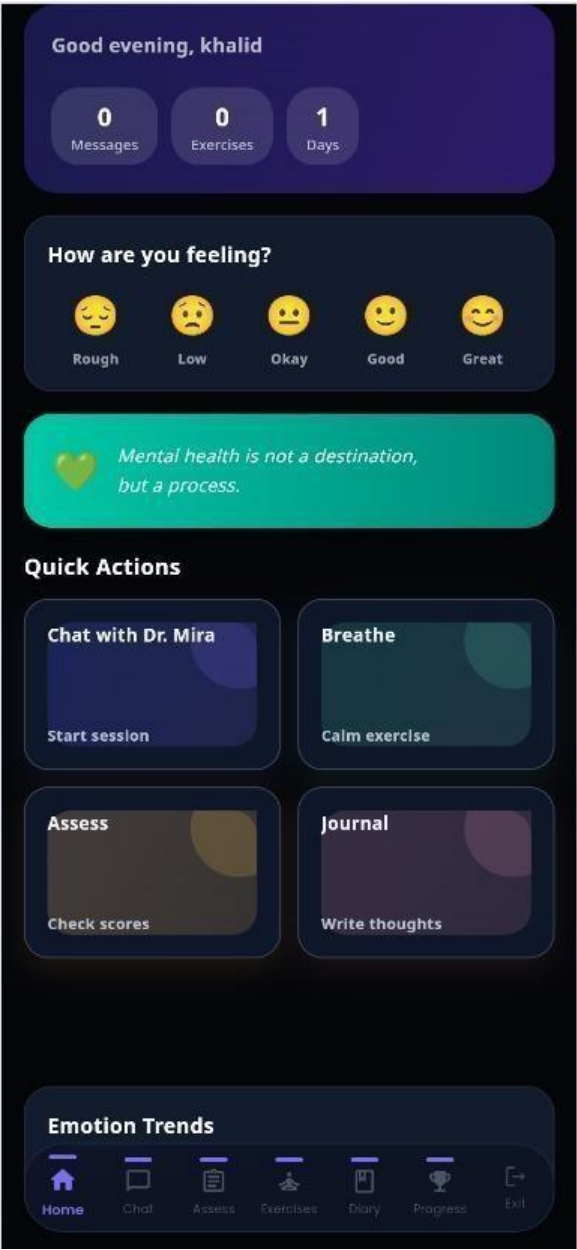


Figure 4.8: Patient Dashboard

Dashboard provides an accessible location for assessments, conversational therapy, interventions, and progress tracking tools.

4.7.3 AI Chat Interface

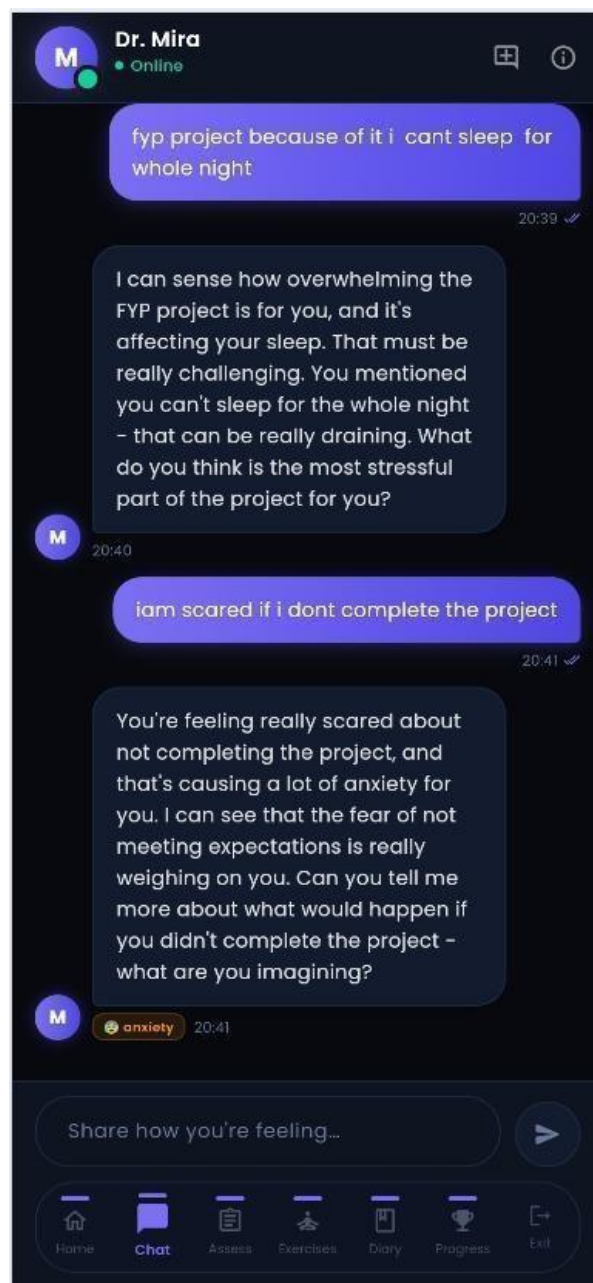


Figure 4.9: AI Conversational Therapy Interface

Chat component helps in interacting with the AI assistant using natural language and getting context-based therapeutic responses.

4.7.4 Clinical Assessment Module

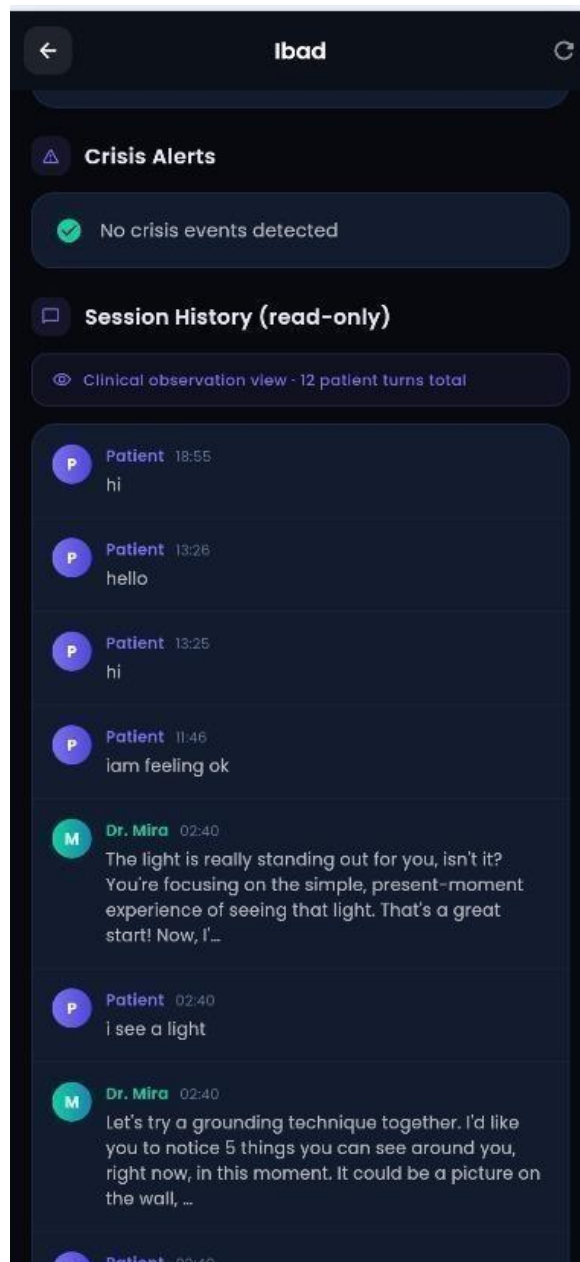


Figure 4.10: Clinical Assessment Screen

Assessment component uses standardized psychological scales and records user responses.

4.7.5 Assessment Results Screen

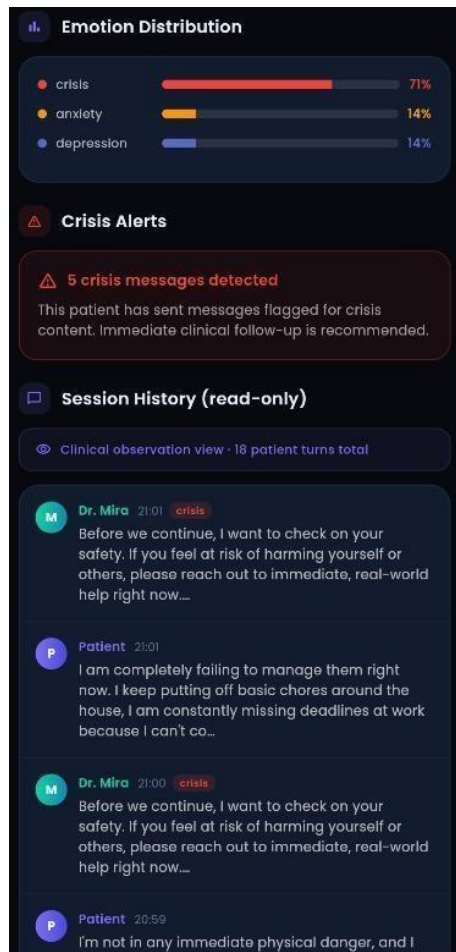


Figure 4.11: Assessment Results and Severity Classification.

Assessment results are displayed using score values, classification severity, and recommendations.

4.7.6 Doctor Dashboard

Clinician dashboard lets healthcare practitioners analyze patient assessments and track progress.



Figure 4.12: Doctor Monitoring Dashboard

4.7.7 Profile and Settings Screen

Profile component allows managing personal information and application settings.

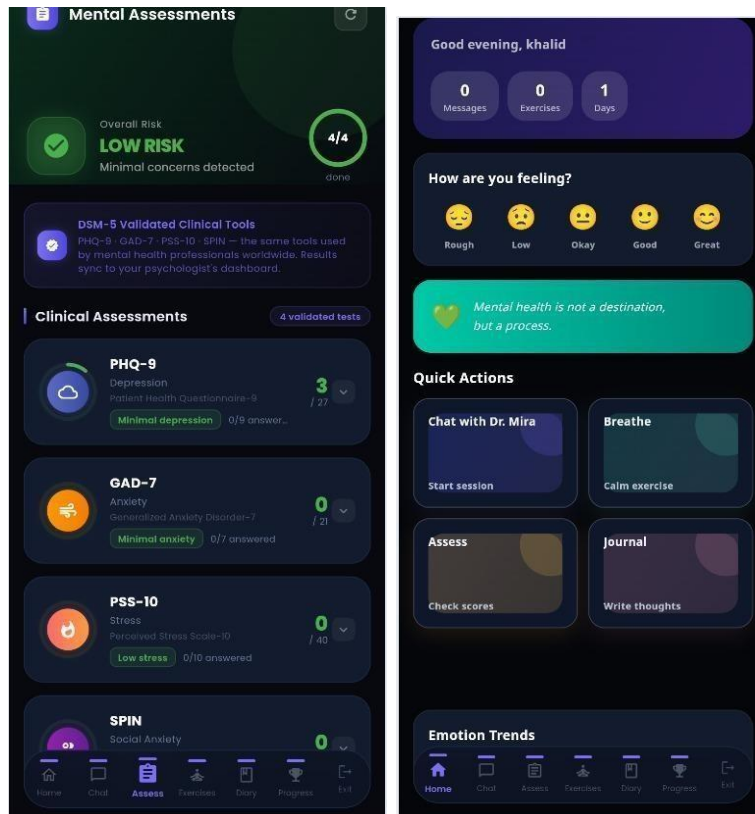


Figure 4.14: User Profile and Settings Interface

4.7.8 Intervention Recommendation Screen

Intervention engine creates personalized cognitive behavioral therapy activities, mindfulness activities, and journaling exercises according to assessment results.

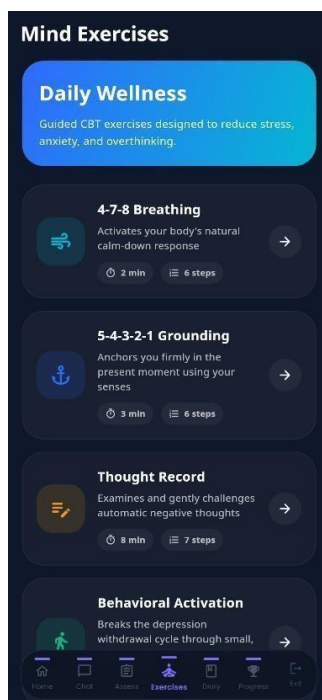


Figure 4.15: Personalized Intervention Recommendations

4.8 LLM + RAG Performance Evaluation

4.8.1 Retrieval Relevance

There were high RAG performance results on information retrieval because this model could always identify relevant information from the knowledge base regarding medicine. The most of the information was close in semantics to the query made by the user. However, there was low efficiency in dealing with emotion-related cultural queries.

4.8.2 Response Clinical Appropriateness

The use of RAG technology has increased the appropriateness of the answers given compared to answers given by large language models. The answers are evidence-based and comply with Cognitive Behavioral Therapy guidelines.

4.8.3 Conversational Quality

The user tests revealed that the created outputs are empathetic and contextual. Conversations that took place in multiple turns remained coherent and showed proper usage of session memory.

4.9 Mobile and Web Application Testing

Both the web app built using Streamlit and the mobile app created using Flutter have been tested for all modules created. The authentication module, scoring module, conversational AI module, progress tracking module, Firebase module, SQLite offline storage module, and the doctor monitoring module have been successfully tested without any major glitches. There were no performance issues with the mobile app.

4.10 System Performance Discussion

Evaluation reveals that OmniMind is able to successfully integrate clinical assessment, artificial intelligence, and mobility into one platform for provision of mental health care services. It is possible by combining standardized psychometric assessments, conversational assessments, fine-tuned large language models, and Retrieval-Augmented Generation to provide evidence-based and context-specific mental health care. There is a lot of potential for implementation of the tool in under-resourced areas because of the possibility of functioning in an offline mode, lightness of the tool, and its mobile-first approach. Nevertheless, there are some weaknesses in the tool, such as limited language choices, lack of real-world validation, and dependence on external artificial intelligence systems. In future studies, there might be developed multilingual solutions, clinical trials with more participants, advanced crisis detection algorithms, and personalization options.

CHAPTER 05

CONCLUSION

5.1 Conclusion

The problems related to mental health disorders have become one of the most acute issues concerning global public health, significantly impacting people of all ages and social groups. However, despite the increasing concern about this problem, access to mental health services is still restricted for many people around the world due to different reasons. In developing countries, shortage of professionals in the domain, stigma related to receiving help from a psychiatrist, financial limitations, and lack of infrastructure in general make it difficult for people to receive proper care for their mental conditions. Thus, there is a need for the development of innovative approaches using artificial intelligence to monitor and treat mental disorders.

This thesis proposes the design, development, and testing of OmniMind—an integrated system of monitoring mental health, which includes real-time emotion detection and adaptive therapeutic interventions. The suggested system combines modern AI technologies and traditional approaches to the assessment of patients' condition to provide an easy-to-use, effective, and personalized tool in the form of mobile application.

The structure of the system includes three layers: a mobile application that was developed using Flutter framework, application logic layer that works on Python technology, and Firebase cloud services used for authentication and data storage. It incorporates popular tools for measuring mental health, such as Patient Health Questionnaire (PHQ-9), Generalized Anxiety Disorder Scale (GAD-7), Perceived Stress Scale (PSS-10), and Social Phobia Inventory (SPIN). All tools are used in accordance with current recommendations and can be applied effectively to produce reliable results and severity level classifications.

One of the key contributions of this research is the usage of the Retrieval-Augmented Generation (RAG) framework with the Large Language Models (LLMs). It allows providing context-aware and clinically validated conversation support through combining semantic knowledge retrieval, emotion recognition, results of the clinical assessment, and conversation history. The implementation of the fine-tuned LLaMA 3.1-8B model with Parameter-Efficient Fine-Tuning (PEFT) and Low-Rank Adaptation (LoRA) improves the possibility to generate empathic and clinically accurate responses following the principles of Cognitive Behavioral Therapy (CBT).

In addition, the platform contains several supporting modules, which include a keyword-based crisis detection system, an adaptive interventions recommendation engine, a progress tracking dashboard, gamification features, and data persistence using SQLite. Together, they create a full-fledged ecosystem for digital monitoring and personal therapeutic support.

The evaluation results described in Chapter 4 show the feasibility and efficiency of the proposed approach. The fine-tuning of the language model proved to be successful, showing fast convergence, as the training loss decreased rapidly within ten epochs. Comparative evaluation revealed that the fine-tuned language model surpassed the baseline model in generating high-quality conversational responses that are appropriate both in terms of context and clinically. Thus, the introduction of the RAG architecture allowed

improving the reliability of generated answers and minimized the possibility of creation of the hallucinated or medically incorrect content.

The functional tests proved the successful implementation of all system elements, including user identification, clinical evaluation, conversational AI model, crisis detection, intervention suggestions, and result visualization. Mobile application tests revealed that all elements function successfully, while Firebase integration allows ensuring data security. The module of clinical evaluation uses appropriate scoring methods and severity classification, which makes it possible to use it as the digital instrument for evaluating mental state.

In other words, the results of testing reveal that OmniMind satisfies the goals that were set initially. The presented system shows how it is possible to combine artificial intelligence, clinical evaluation, and mobile technologies to develop an accessible and personalized instrument for supporting mental health. Even though the system does not substitute professional psychiatrist in diagnosing and treating disorders, it is a significant achievement in the field of mental health care.

5.2 Research Contributions

Key contributions from the research work can be listed as follows:

1. Building of a holistic AI-powered framework for monitoring of mental health that consists of assessment, intervention, and conversational support components.
2. Embedding of clinically validated tests of psychological assessment (PHQ-9, GAD-7, PSS-10, and SPIN) in a mobile health app.
3. Building of the Retrieval Augmented Generation (RAG) system for delivering conversational assistance on a clinically sound basis.
4. Fine-tuning of the LLaMA 3.1-8B large language model with LoRA (parameter-efficient tuning method) to deliver mental health-related dialogues.
5. Building of an adaptive engine for generating recommendations for interventions based on assessment results, users' emotions, and interaction history.
6. Designing of the crisis detection algorithm for detecting dangerous language patterns and triggering corresponding safety measures.
7. Implementation of the progress tracking and gamification framework to increase users' engagement.
8. Development of the mobile-first solution that can be easily deployed in low-resource settings due to offline mode and light design.

5.3 Limitations

Despite the success of OmniMind, there are still some limitations that need to be taken into account.

5.3.1 Limited User Evaluation

The system was tested on a fairly limited number of users. This means that the outcomes regarding usability and engagement cannot be applied on a wider scale.

5.3.2 Limited Knowledge Base Coverage

Effectiveness of the RAG method depends on how vast and good is the knowledge base. Though the knowledge database contains CBT information, psychoeducational information, and mental health information, there may be some unique or rare cases in psychology which will be not covered by the knowledge base.

5.3.3 Language Constraints

At the moment, the system supports English communication only. Individuals communicating in Urdu,

Sindhi, and other regional languages face possible restrictions related to accessibility and communication quality.

5.3.4 Rule-Based Crisis Detection

The crisis detection module uses rule-based matching algorithms for crisis detection, which might be insufficient for detecting psychological problems expressed indirectly or culturally.

5.3.5 Dependency on AI Services from External Cloud Providers

Specific functions of the AI rely on cloud-based API access and internet connection. Availability, service latency, and other constraints related to APIs might influence the performance of the system.

5.3.6 No Formal Clinical Trial

Despite the fact that the assessment tools were validated against clinical standards and recommendations, the therapy provided by OmniMind was not subjected to formal clinical trials or randomized controlled experiments.

5.3.7 Non-Diagnosis Character of the Therapy

OmniMind provides only auxiliary and supporting service and does not provide formal diagnoses, prescription of treatment, and substitute professional mental healthcare services.

5.4 Future Work

There are many possibilities to develop OmniMind further.

5.4.1 Multilingual Support

In future, the system should support Urdu, Sindhi, and other regional languages. Multilingual NLP models will greatly increase the accessibility of the system in Pakistan and other multilingual communities.

5.4.2 Multimodal Emotion Detection

In future, the research can integrate multimodal emotion recognition technology using facial expressions, voice, and physiological signals together with textual data. This will increase accuracy of emotion estimation.

5.4.3 Better Crisis Detection

Machine learning and deep learning methods can be used instead of keyword-based crisis detection algorithm. More context-aware risk prediction models might help detect problems more efficiently.

5.4.4 Clinician Dashboard Development

Fully-developed clinician dashboard can allow licensed mental health professionals to track patients' progress, observe risks, perform remote supervision and intervention without violation of patients' privacy and consent.

5.4.5 Wearable Devices Integration

Integrating OmniMind with smartwatches and other wearable sensors can allow to monitor patients' physiological indicators like heart rate variability, sleeping patterns, physical activities, and others.

5.4.6 Clinical Validation with Big Samples

Further research can include large-scale clinical trials with participation of various patients to estimate usability and engagement of the system in real-life situation.

5.4.7 RCTs

Clinical trials involving patients in comparison with control group will help to determine the effectiveness of the therapy better.

5.4.8 Continuous Model Improvement

Further improvements can include the search for larger datasets in the specific domain, application of reinforcement learning, advanced retrieval techniques, and new language model architectures.

5.4.9 Deployment and Commercialization

Future research can concentrate on making OmniMind production-ready healthcare solution by means of cloud deployment, licensing, healthcare partnership, and involvement in national digital health projects.

5.5 Final Remarks

The swift advancement of AI technology has brought about tremendous possibilities for expanding the provision of mental health assistance. Through the integration of clinical evaluation techniques, Retrieval-Augmented Generation, large language models, and intelligent adaptive intervention, OmniMind proves how intelligent digital tools can help deliver mental health assistance on scale.

While there are numerous challenges in making the technology widely adopted clinically, multilingually accessible, and extensively validated, the results of this study show that AI-driven mental health platforms can indeed assist people in maintaining their emotional balance and detecting psychological problems earlier.

OmniMind is a remarkable attempt at building an intelligent and user-friendly digital mental health ecosystem, which could cater to the increasing demand for mental health assistance all around the world.

The following limitations of the research have to be taken into account:

Sample size of user testing: the preliminary testing of the tool was done using a rather small sample size and therefore lacks statistical significance. Controlled trials need to be conducted in order to evaluate the engagement and usability of the application properly.

RAG Knowledge Base Scope: although the RAG knowledge base used in the project has been clinically verified, it still has certain limits in terms of the number of possible scenarios of users' queries. Some of the edge cases—especially highly idiomatic forms of distressing thoughts and culturally-specific idioms in Pakistani vernacular—might be beyond the scope of the knowledge base and thus result in inadequate retrieval.

Language Support: although the current version of OmniMind supports only English-language user input, many people in Pakistan use other languages when expressing their ideas and emotions, namely Urdu, Sindhi, and others.

No RCT Done: the application has never undergone a Randomized Controlled Trial, which is the most reliable way of measuring the clinical effectiveness of any medical treatment.

Only a Supplementary Tool: as it has already been said, OmniMind is just a supplementary mental health platform and cannot be used as a clinical treatment itself.

5.6 Future Directions

The below-mentioned directions have been put forth for further research and development:

Support for Urdu and Sindhi Languages: Supporting Urdu and Sindhi languages through OmniMind NLP pipeline will enable the system to target the largest number of people in Pakistan as these are the two most widely spoken languages in the country.

Therapist Dashboard: Full integration of the therapist dashboard which will enable verified psychologists to access anonymized information about user's progress and send out any high risk notifications or remote advice would enable the system to become an important part of the healthcare system.

Computer Vision Module: The system would be enhanced by incorporating the computer vision module which would analyze facial expressions of the patients in order to recognize the inconsistencies between emotions and their expressions.

Randomized Controlled Trial: The implementation of a randomized controlled trial to compare the mental health outcomes of OmniMind users and non-users (waitlist or psychoeducation only) would give a necessary evidence base for using the tool in formal health setting.

Wearable Integration: Future iterations may use consumer grade wearable devices (smartwatch) in order to incorporate physiological data (HRV, sleep, and physical activity) as inputs for estimating user's emotional state according to the CARESS model discussed in literature.

Publication: Publishing the results of the project in the conference or journal dealing with digital health, medical AI, and human-computer interaction.

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